

NOVA
IMS
Information Management School

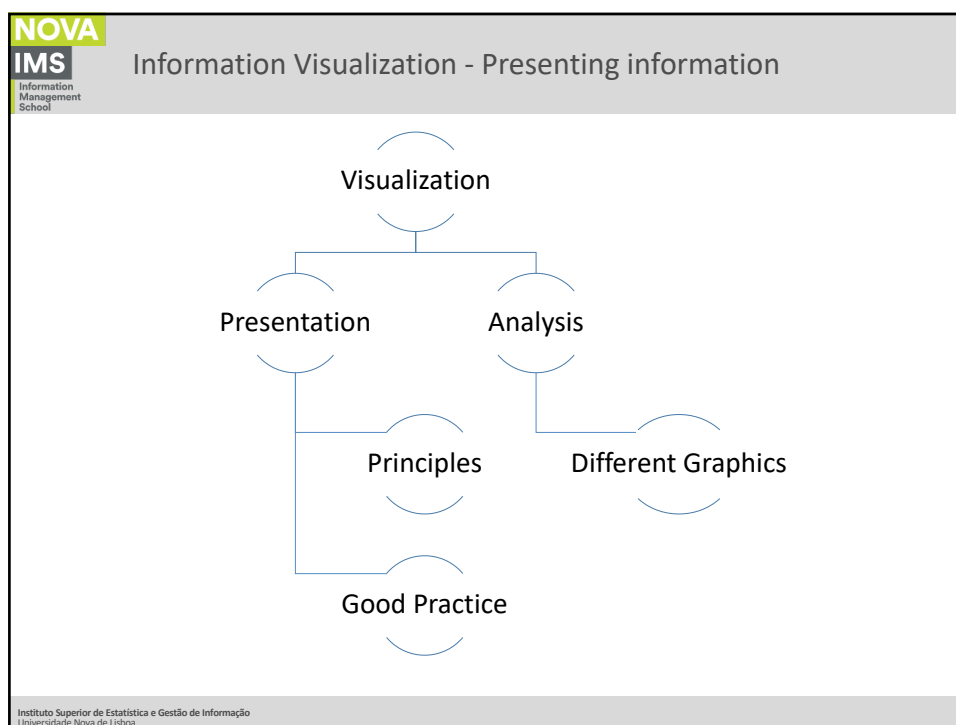
Data Mining I

S3

NOVA-IMS 2024/2025
Fernando Lucas Bação
bacao@isegi.unl.pt
<http://www.isegi.unl.pt/fbacao>

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

1



2

[illegible]

3

NOVA
IMS
Information
Management
School

Information Visualization – 5 Influential

Napoleon's 1812 Russian Campaign

Legend:
 ● causes
 ● preparations

Map showing the invasion and retreat routes of Napoleon's 1812 Russian Campaign. The map includes troop numbers at various stages and locations, and a temperature scale in Celsius.

Troop Numbers:

- At the start: 427,000
- At Minsk: 10,000
- At the first battle: 4,000
- At the second battle: 8,000
- At the third battle: 12,000
- At the fourth battle: 28,000
- At the fifth battle: 30,000
- At the sixth battle: 20,000
- At the seventh battle: 24,000
- At the eighth battle: 37,000
- At the ninth battle: 35,000
- At the tenth battle: 87,000
- At the eleventh battle: 94,000
- At the twelfth battle: 127,100
- At the thirteenth battle: 100,000
- At the fourteenth battle: 100,000

Locations: Minsk, Polotsk, Smolensk, Dnieper, Borodino, Malo-Yaroslavets, Moscow.

Temperature in Celsius:

- 33°
- 38°
- 30°
- 14°
- 26°
- 11°
- 0°
- 13°
- 25°
- 38°

Scale: 100 miles

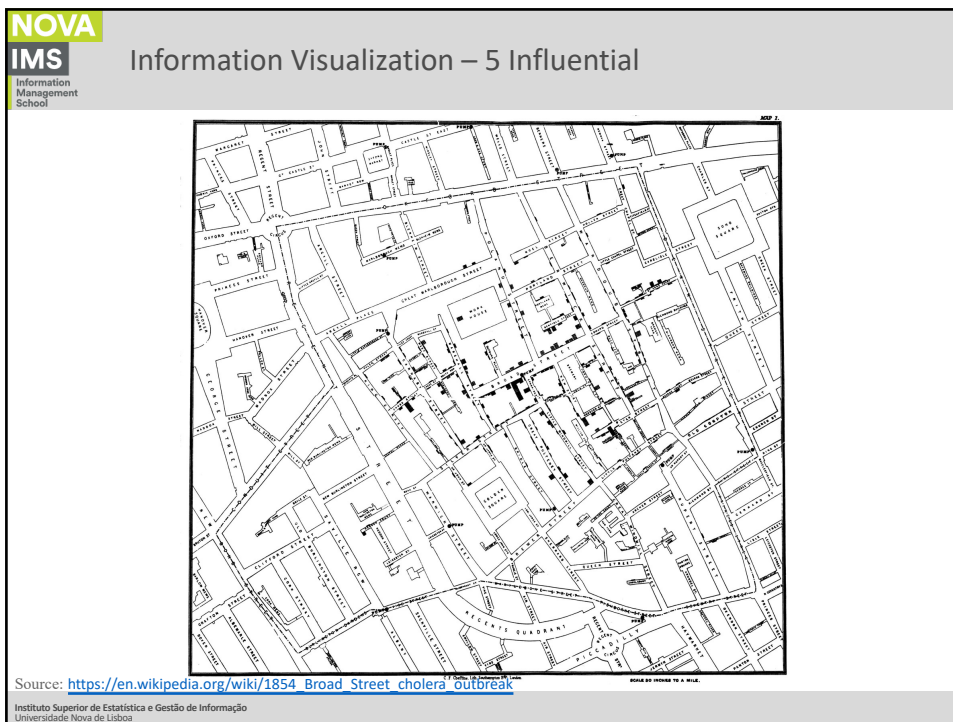
References:

Source: https://www.d.umn.edu/~cstroupe/ideas/representational_space_11.html

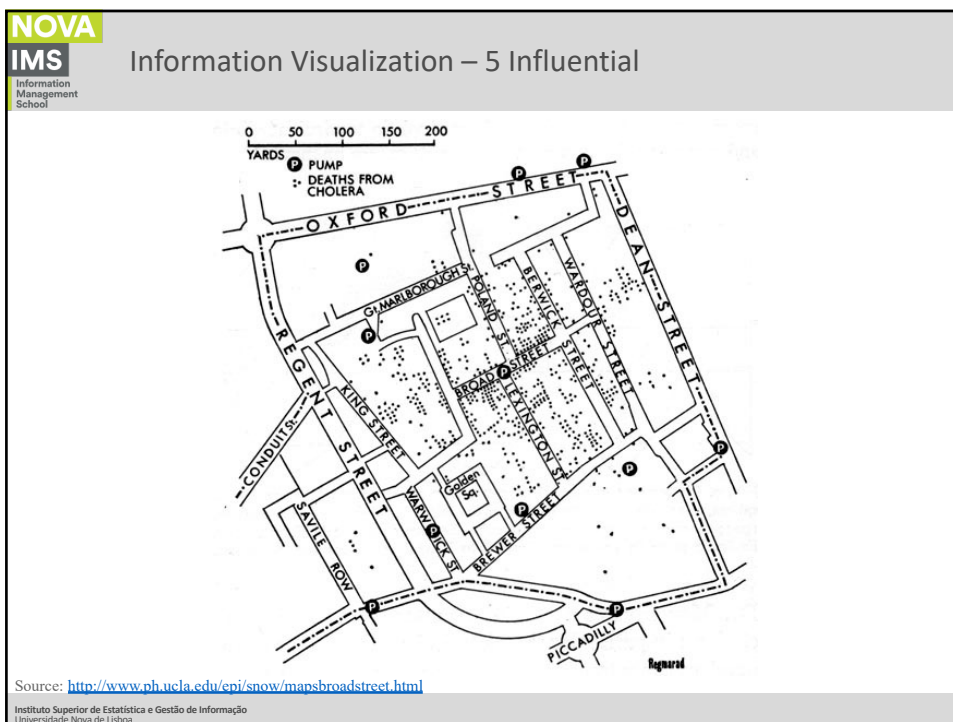
Instituto Superior de Estatística e Gestão de Informação

Universidade Nova de Lisboa

4



5



6

NOVA
IMS
Information Management School

Information Visualization Strong Points

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

UNICIS

ACES

Schools eduniversal

7

NOVA
IMS
Information Management School

Information Visualization – Strong Points

EGG CURRY

4 PEOPLE, MAIN DISH
SERVED WITH WHITE RICE & RED BEANS

```

graph LR
    A[8 eggs] --> B[boil 10']
    B --> C[stir 10']
    C --> D[halve]
    D --> E[utensils]
    F[2 onions] --> G[grind]
    G --> H[thyme]
    I[2 garlic cloves] --> J[grind]
    K[1 piece ginger] --> J
    L[1 hot pepper] --> J
    M[salt] --> J
    N[pepper] --> J
    J --> H
    H --> C
    O[4 ripe tomatoes] --> P[chop]
    P --> C
    Q[safron] --> C
    C --> R[simmer 5']
    R --> S[stir 10']
    S --> D
    
```

EGGPLANT ROUGAIL

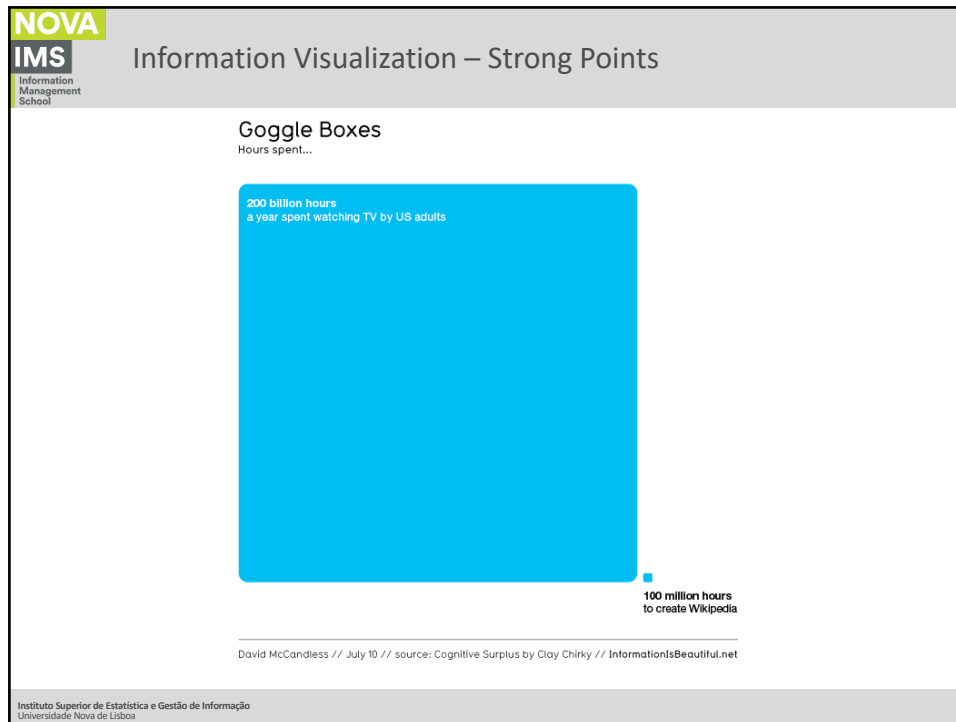
4 PEOPLE, SIDE DISH
SERVED HOT OR COLD. GOES WELL WITH MEAT OR EGGS.

```

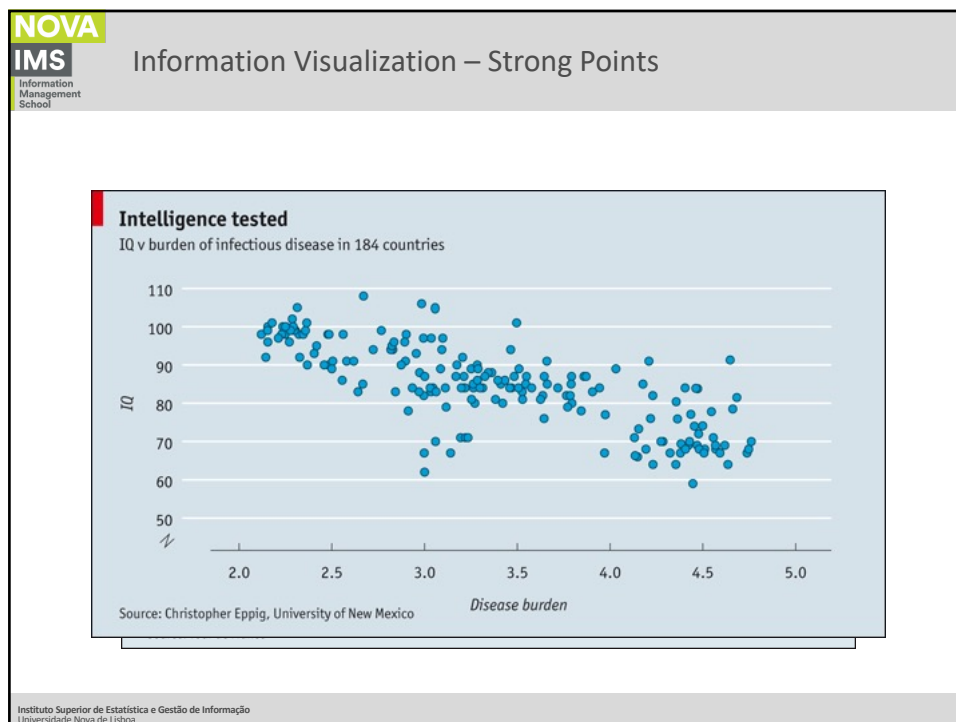
graph LR
    A[2 eggplants] --> B[boil until soft]
    B --> C[scoop flesh, discard skin]
    C --> D[mash]
    D --> E[4 teaspoons of oil]
    E --> F[utensils]
    G[2 white onions or shallots] --> H[grind]
    I[2 hot peppers] --> J[grind]
    K[onion & pepper mix] --> H
    H --> E
    
```

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

8



9



10

NOVA

IMS

Information
Management
School

Information Visualization – Strong Points

Beer consumption
Latest year available

100 Hectolitres, m
(00) Litres per person

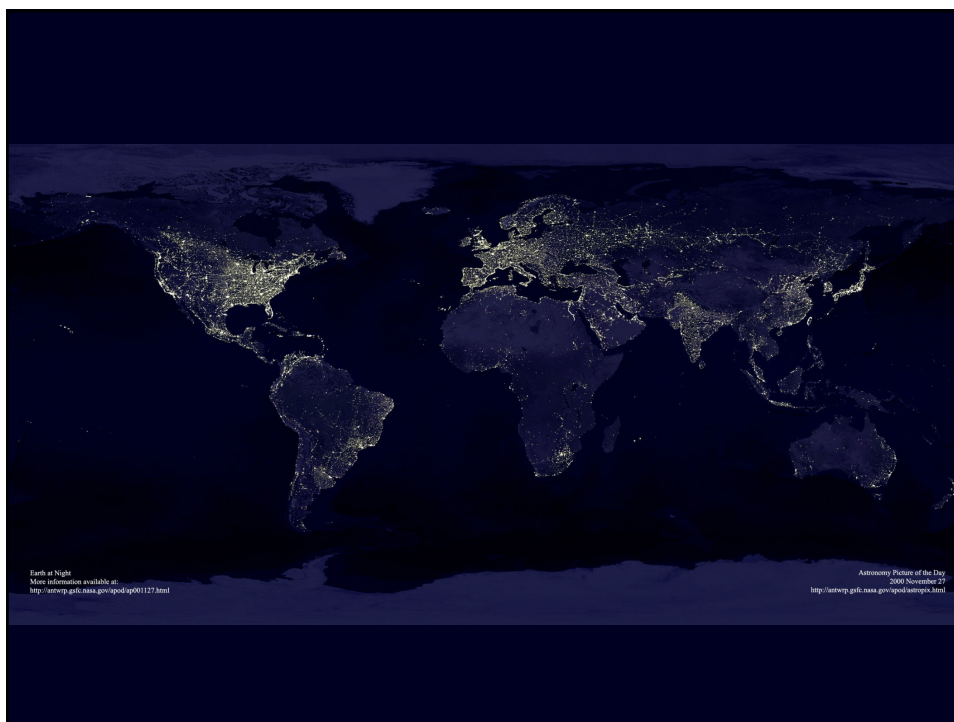
Beer production
Hectolitres, m

Year	Asia	Europe
2008	~550	~550
2009	~550	~550

Sources: Credit Suisse; Canadean; Kirin Institute of Food and Lifestyle

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

11



12

NOVA
IMS
Information Management School

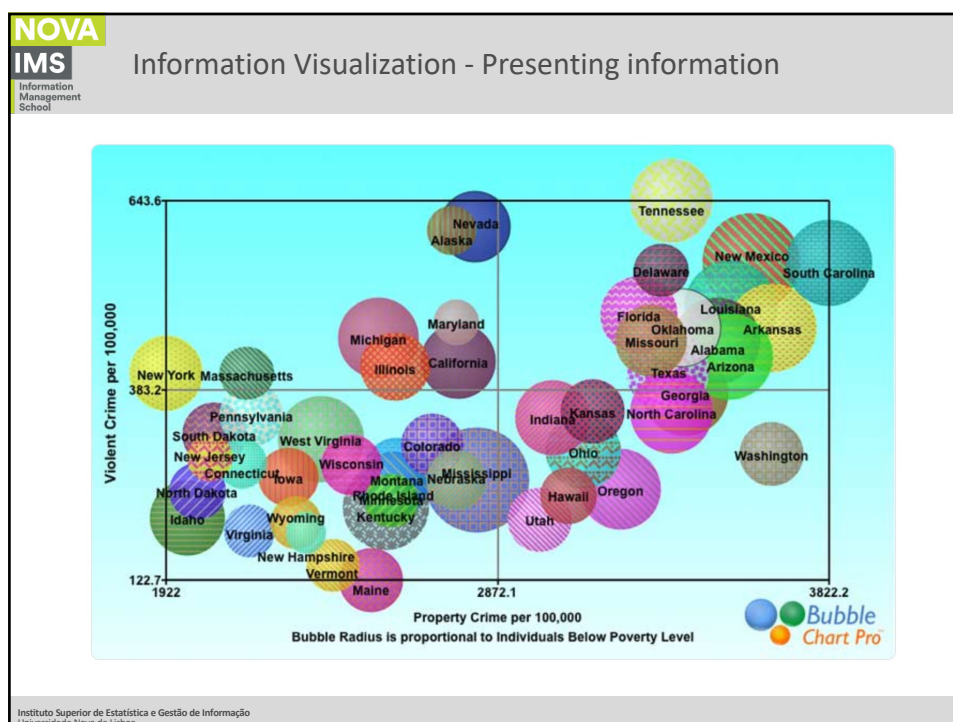
Information Visualization

Please, don't do this!!!

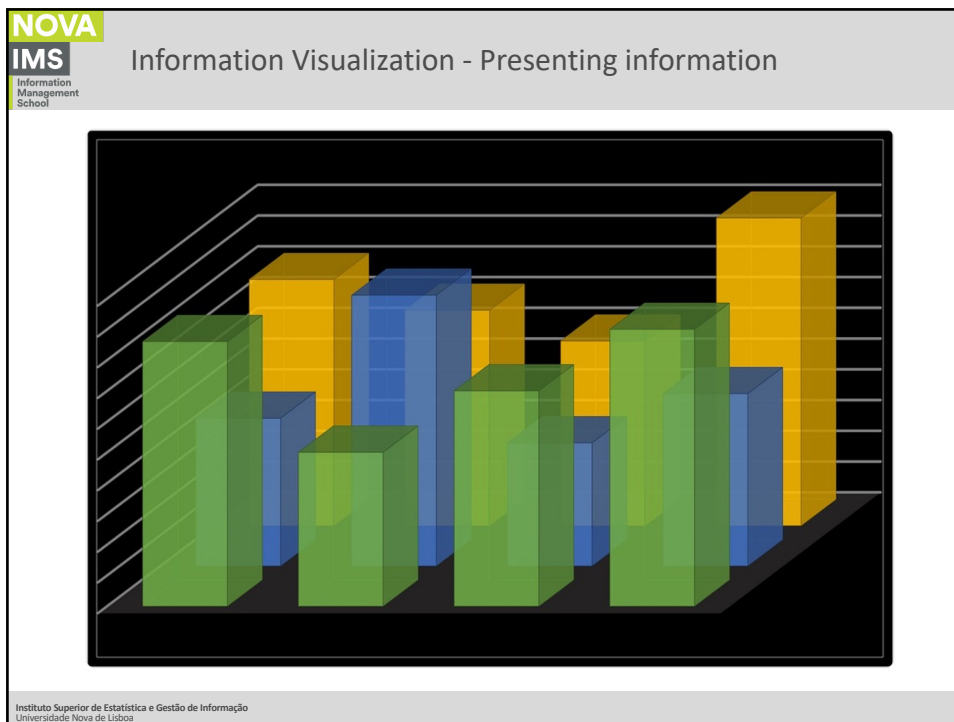
Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

UNICIS

13



14



15



16

NOVA
IMS
Information Management School

Information Visualization

Some guidelines

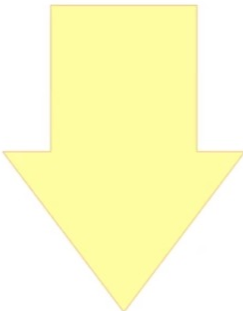
Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

17

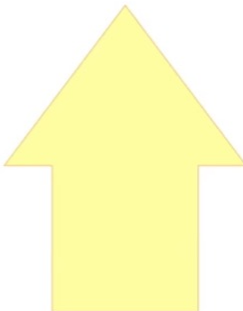
NOVA
IMS
Information Management School

Information Visualization - Presenting information

Fundamental rule of chart design



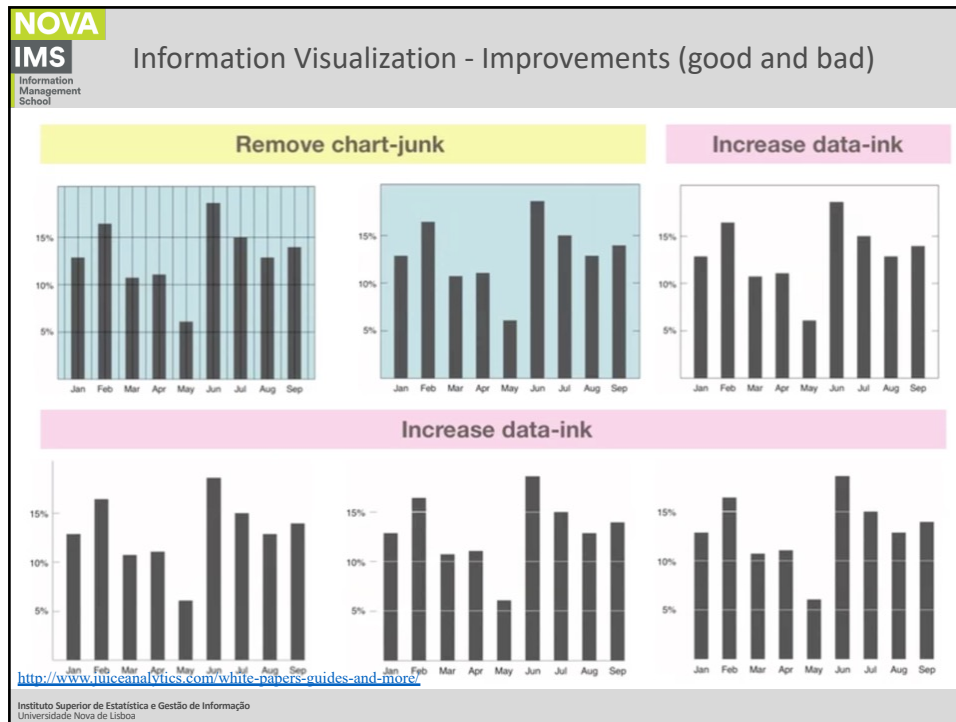
Reduce *chartjunk*
(remove chart elements that are decorative or ornamental)
<http://www.iuiceanalytics.com/white-papers-guides-and-more/>



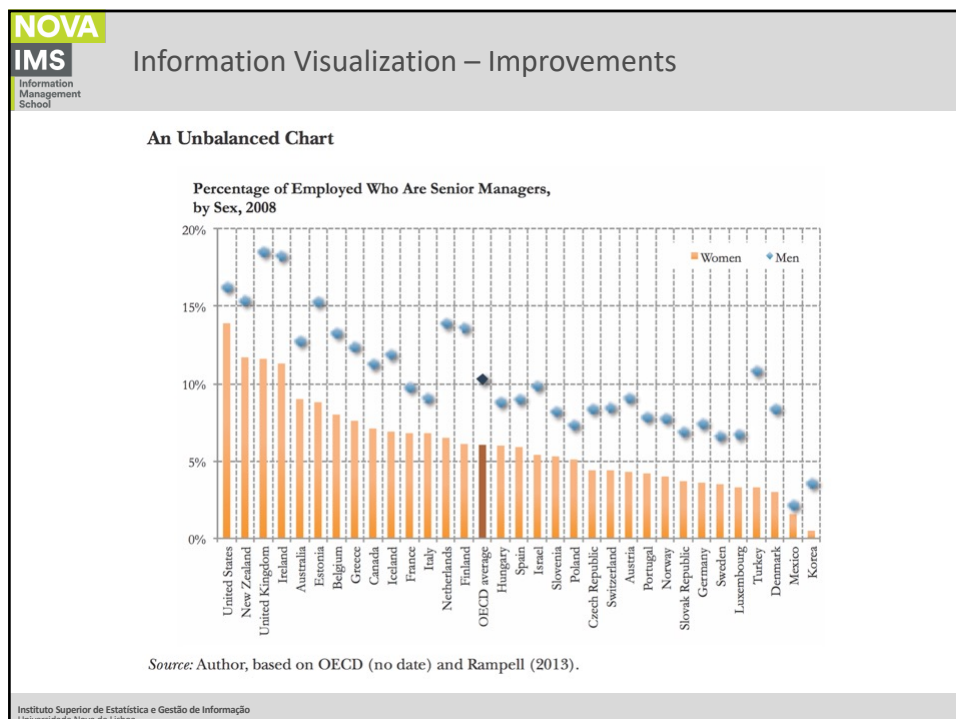
Increase *data-ink ratio*
(make every pixel tell a story about your data)

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

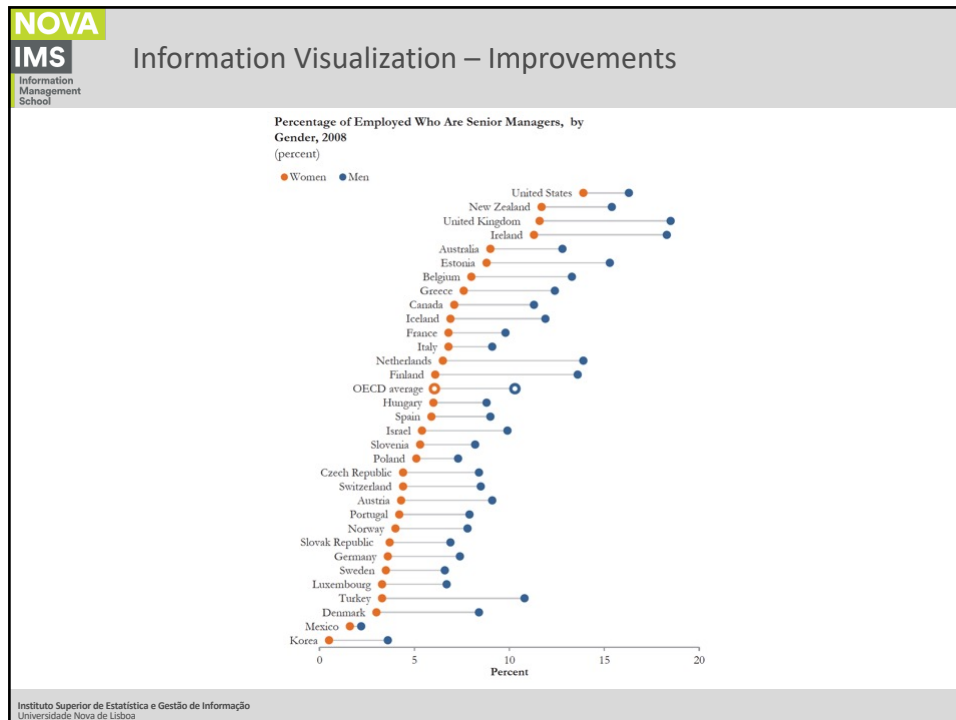
18



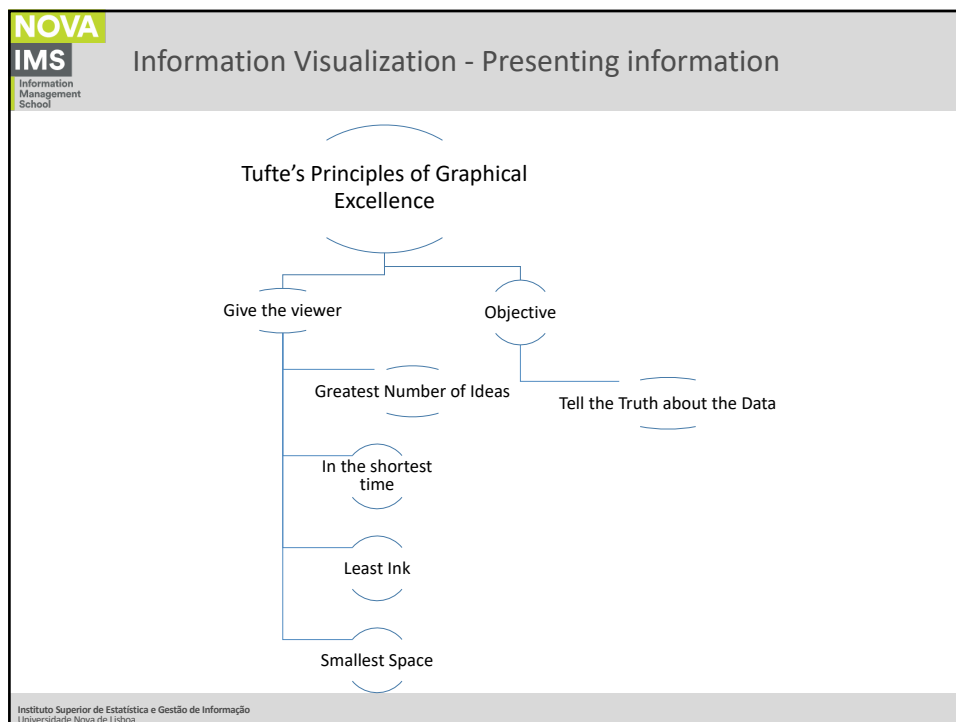
19



20



21



22

NOVA
IMS
Information Management School

Information Visualization Tufte Lie factor

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

UNICIS

ACES

Schools eduniversal

23

NOVA
IMS
Information Management School

Information Visualization - Tufte Lie factor

This line, representing 18 miles per gallon in 1978, is 0.6 inches long.

This line, representing 27.5 miles per gallon in 1985, is 5.3 inches long.

Fuel Economy Standards for Autos
Set by Congress and supplemented by the Transportation Department. In miles per gallon.

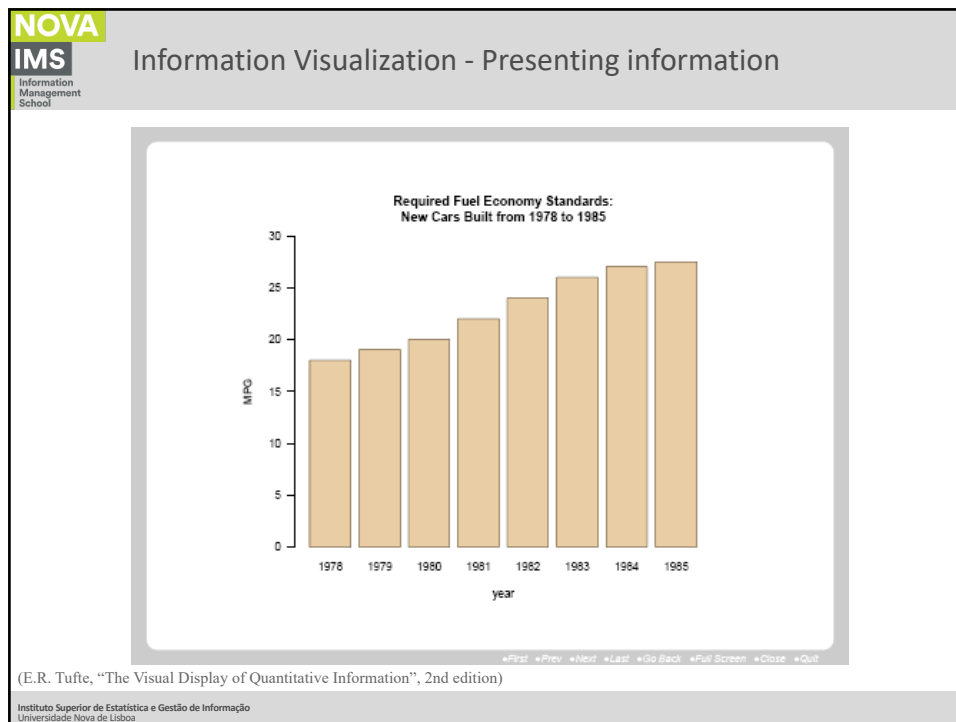
18 19 20 22 24 26 27 27 1/2

New York Times, August 9, 1978, p. D-2.

(E.R. Tufte, "The Visual Display of Quantitative Information", 2nd edition)

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

24



25

NOVA
IMS
Information Management School

Information Visualization - Tufte Lie factor

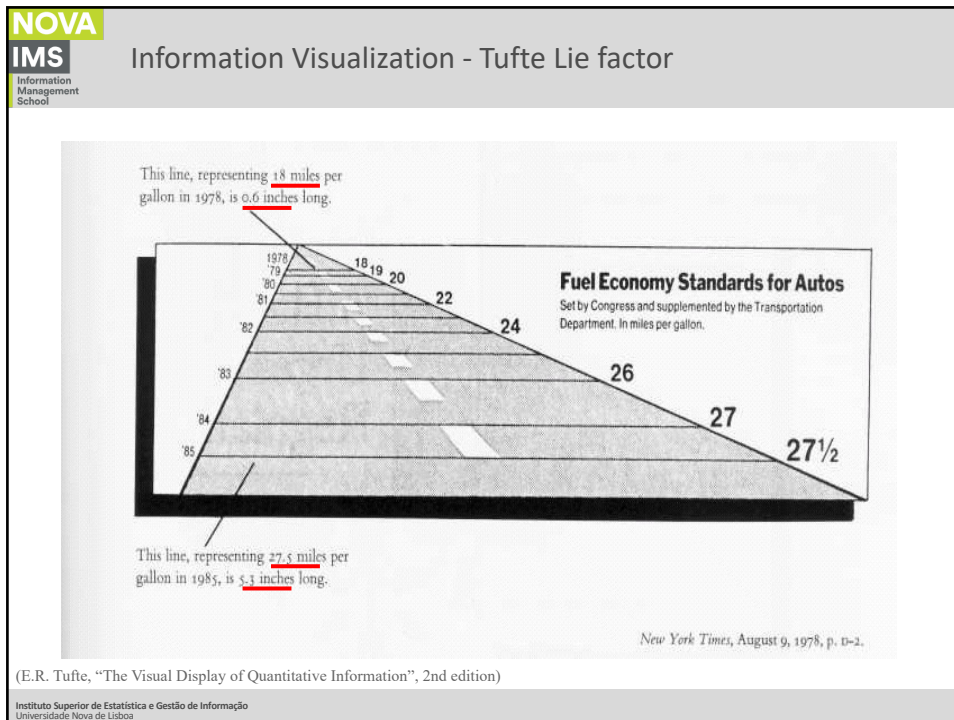
- Ed Tufte of Yale University has defined a "lie factor" as a measure of the amount of distortion in a graph.

$$\text{Lie Factor} = \frac{\text{size of effect shown in graphic}}{\text{size of effect in data}} =$$

- If the lie factor of a graph is greater than 1, the graph is exaggerating the size of the effect.
- Tufte requirement: $0.95 < \text{Lie Factor} < 1.05$

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

26



27

NOVA
IMS
Information Management School

Information Visualization - Tufte Lie factor

$$\text{Lie Factor} = \frac{\text{size of effect shown in graphic}}{\text{size of effect in data}} =$$

$$= \frac{\frac{(5.3 - 0.6)}{0.6}}{\frac{(27.5 - 18.0)}{18}} = \frac{7.833}{0.528} = 14.8$$

Tufte requirement: $0.95 < \text{Lie Factor} < 1.05$

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

28

NOVA
IMS
Information Management School

Information Visualization Improvements (good and bad)

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

UNIS
A3ES
Schools
eduniversal

29

NOVA
IMS
Information Management School

Information Visualization - Improvements (good and bad)

Bad

Month	Sales
Jan	10400
Feb	10600
Mar	10650
Apr	10500
May	10750
June	10700
July	10800
Aug	10700
Sept	10750
Oct	10850
Nov	11050
Dec	11250

Good

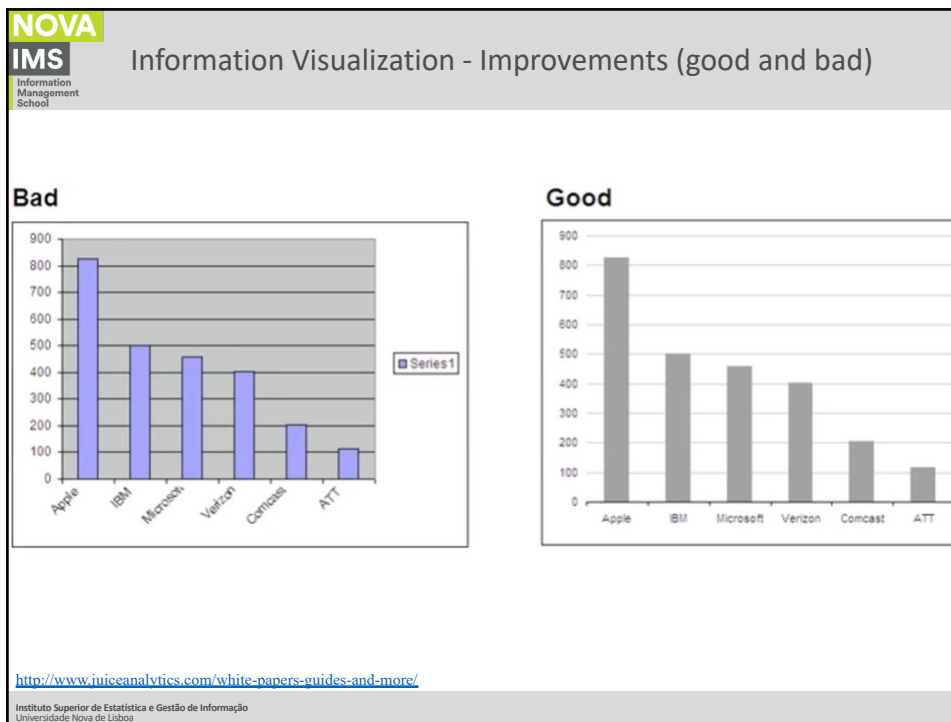
Month	Sales
Jan	10400
Feb	10600
Mar	10650
Apr	10500
May	10750
June	10700
July	10800
Aug	10700
Sept	10750
Oct	10850
Nov	11050
Dec	11250

Maximize the contrast between your data and the background.

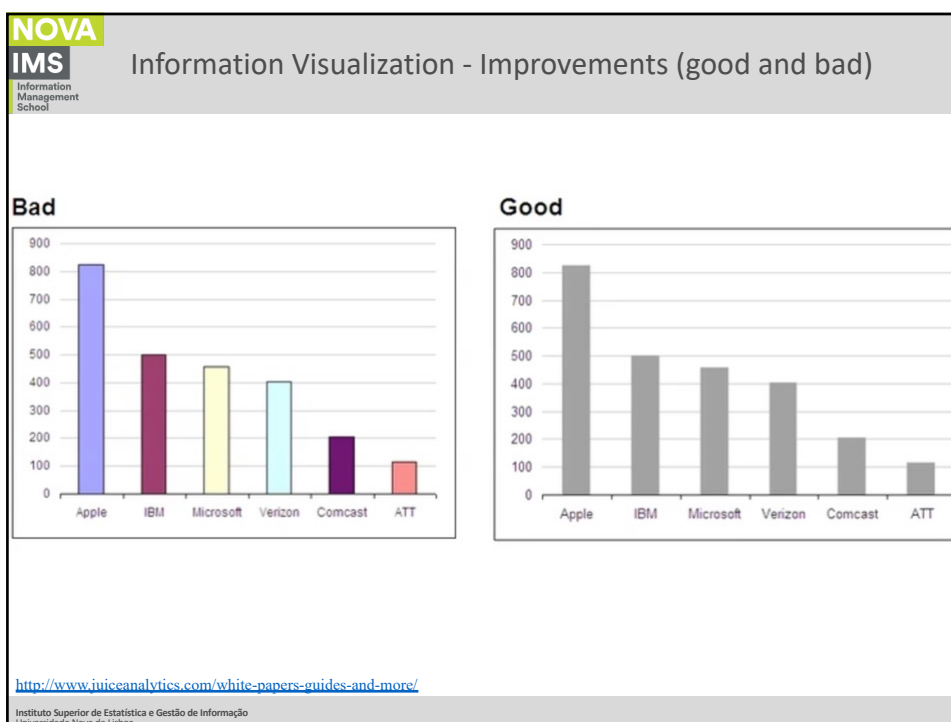
<http://www.iuiceanalytics.com/white-papers-guides-and-more/>

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

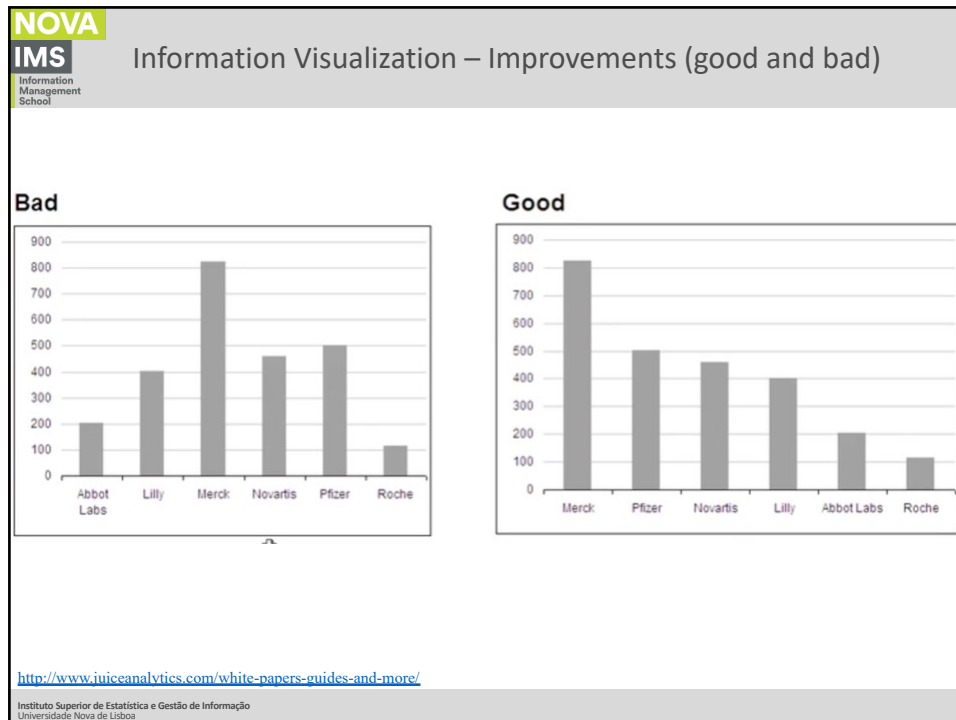
30



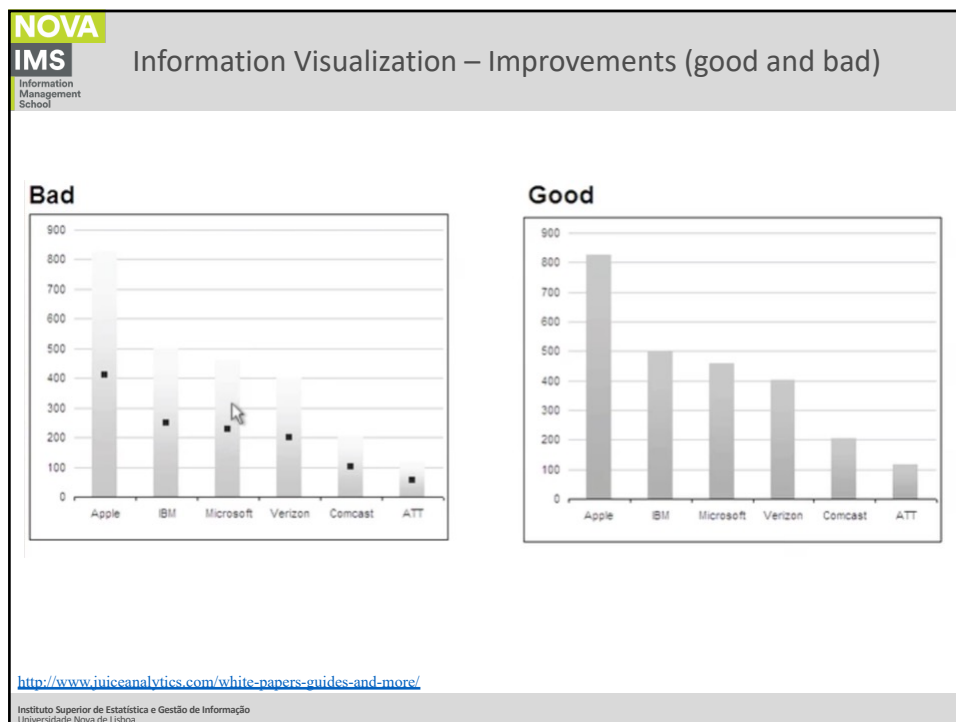
31



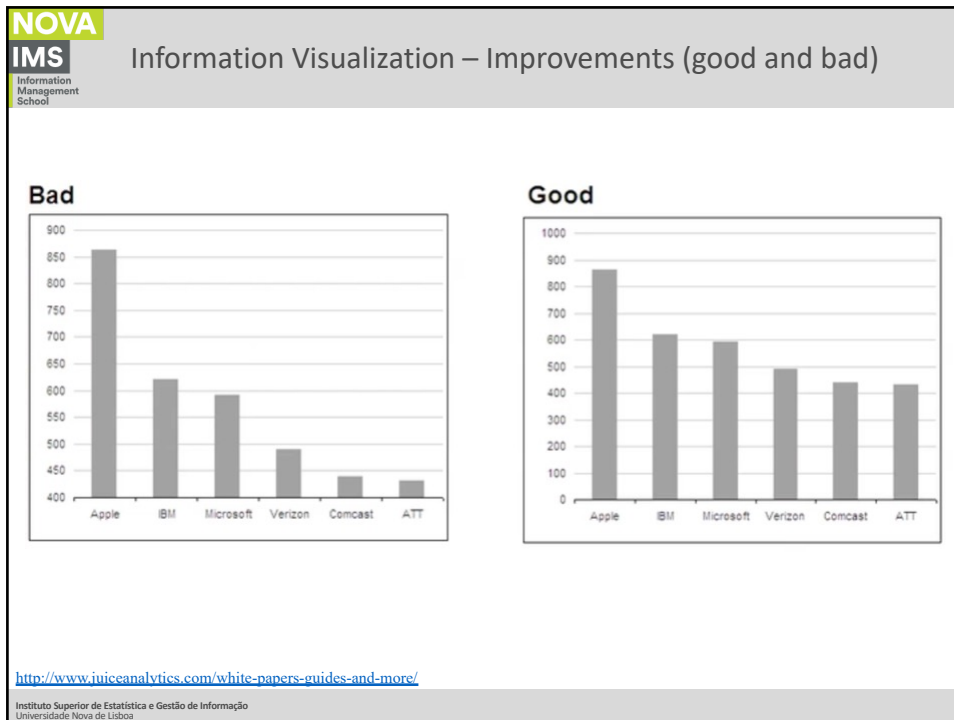
32



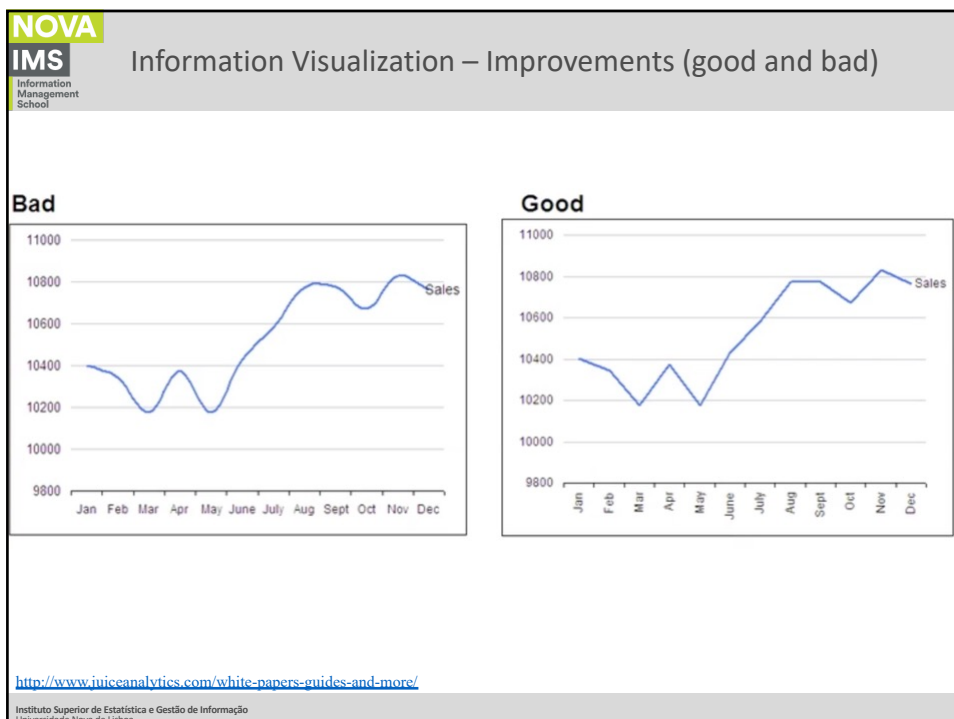
33



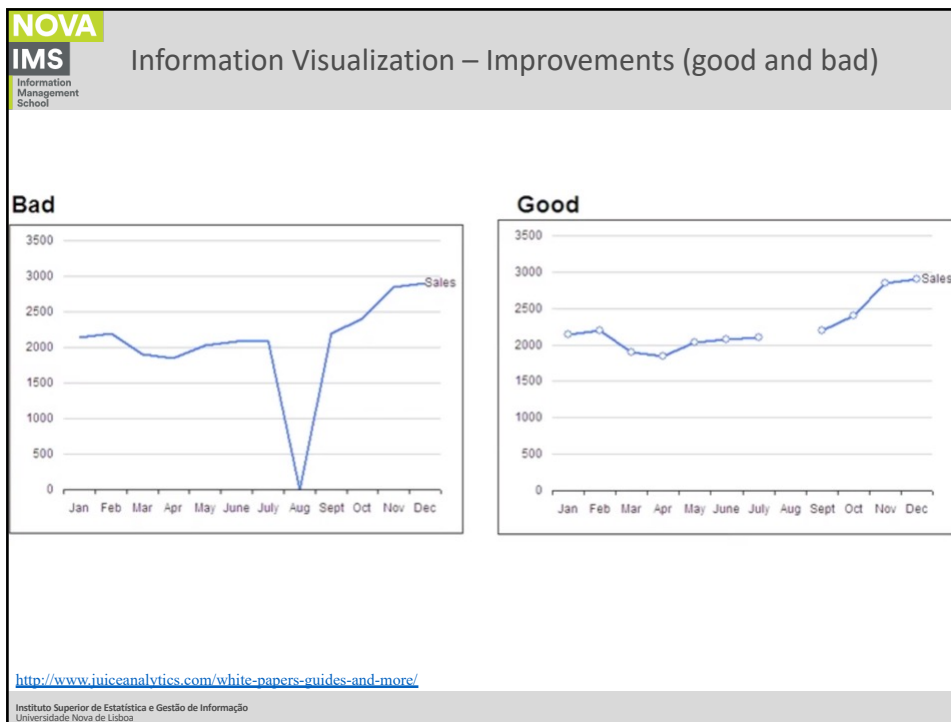
34



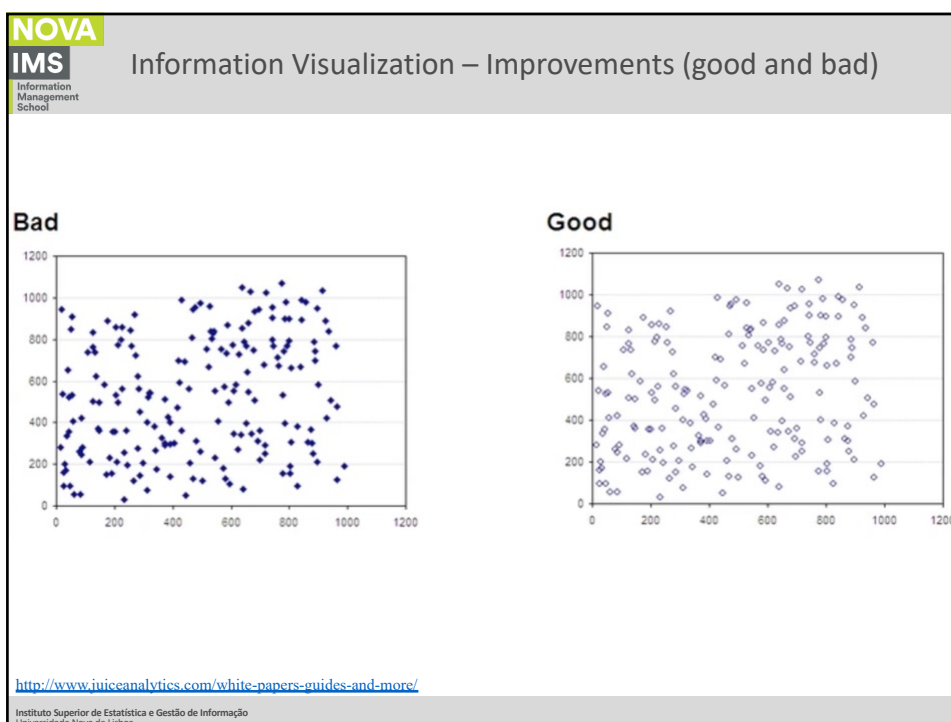
35



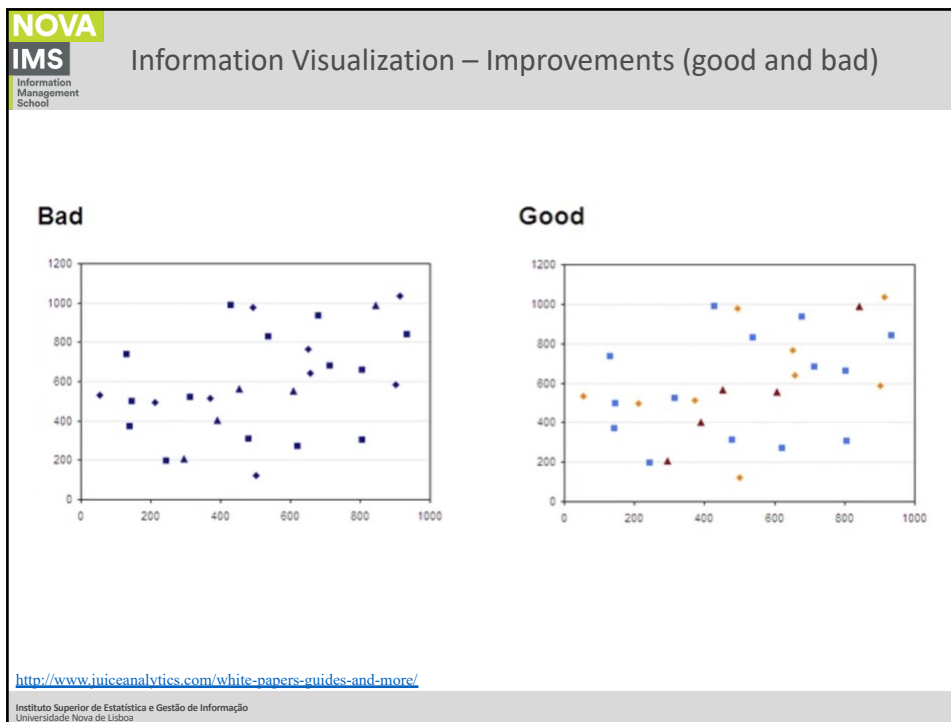
36



37



38



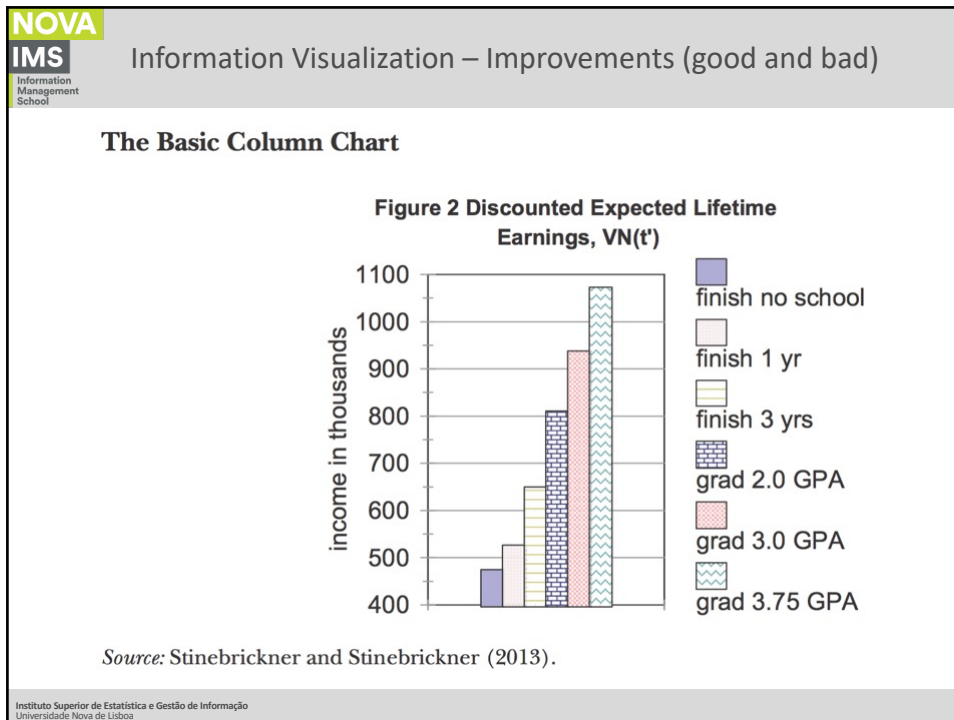
39

NOVA
IMS
Information Management School

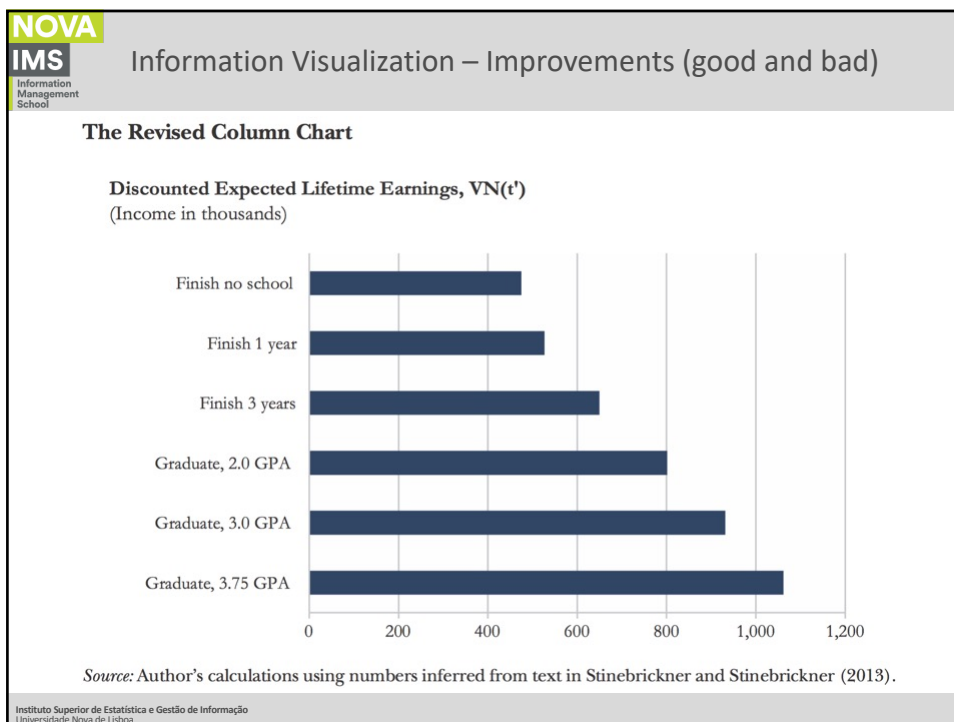
Information Visualization Presenting Information

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

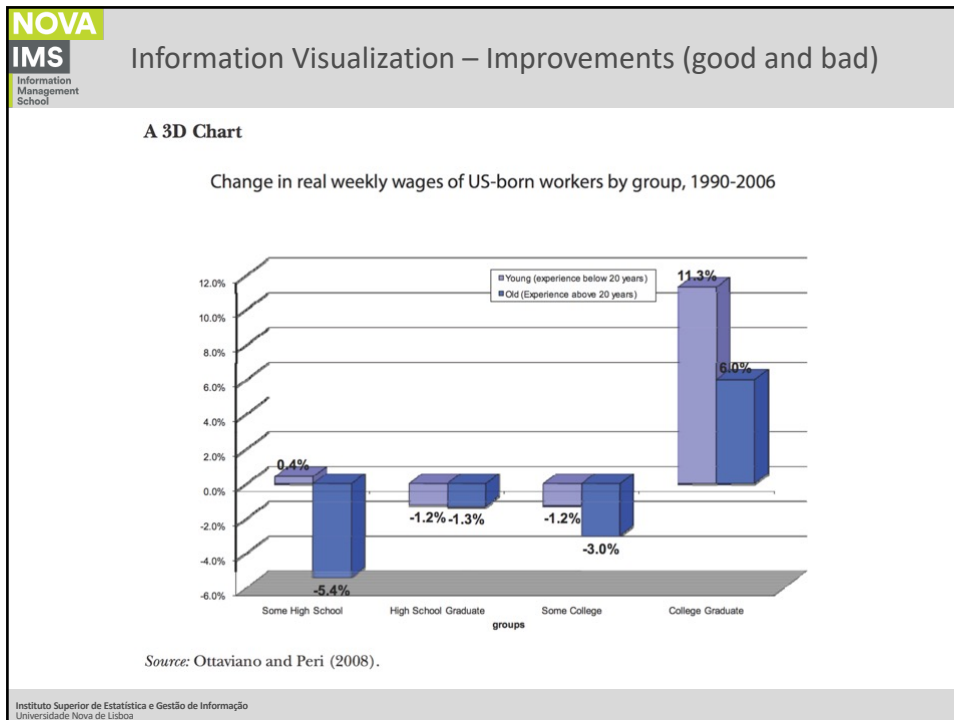
40



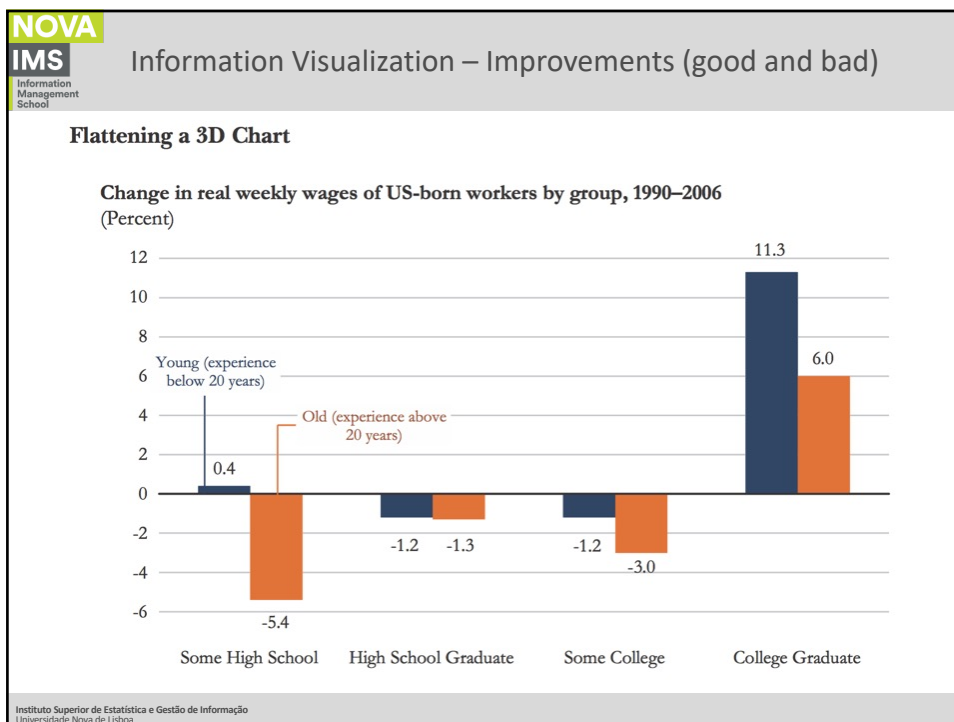
41



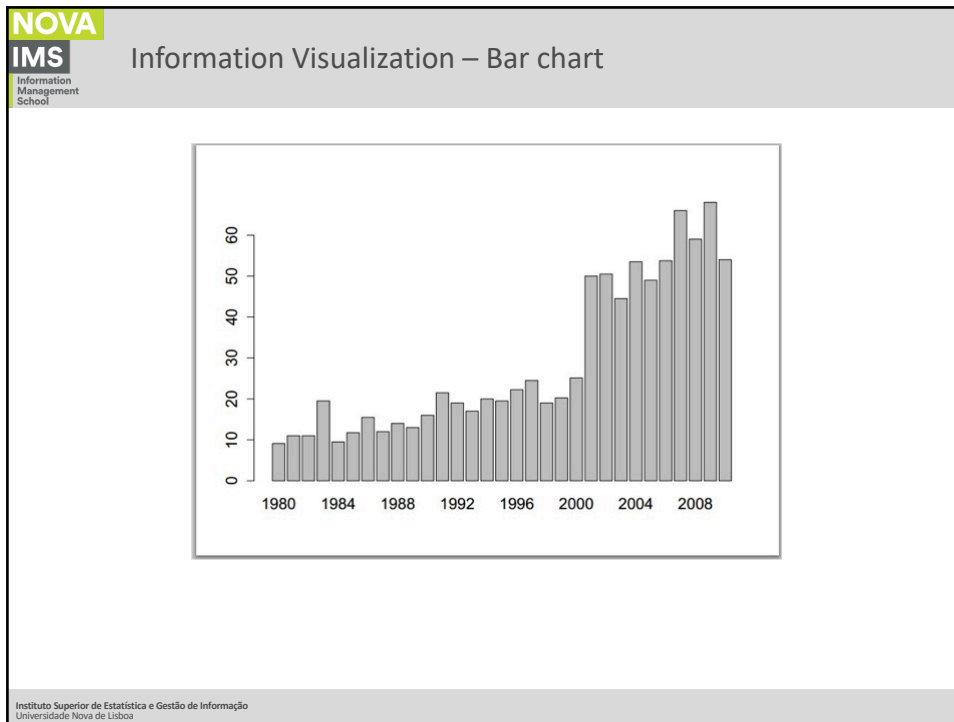
42



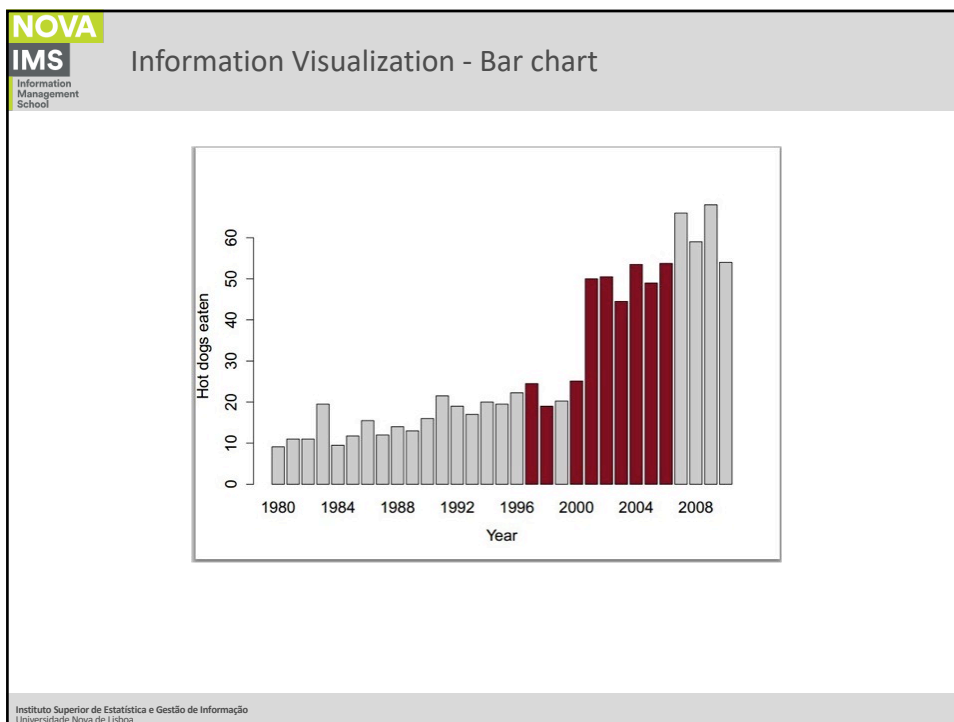
43



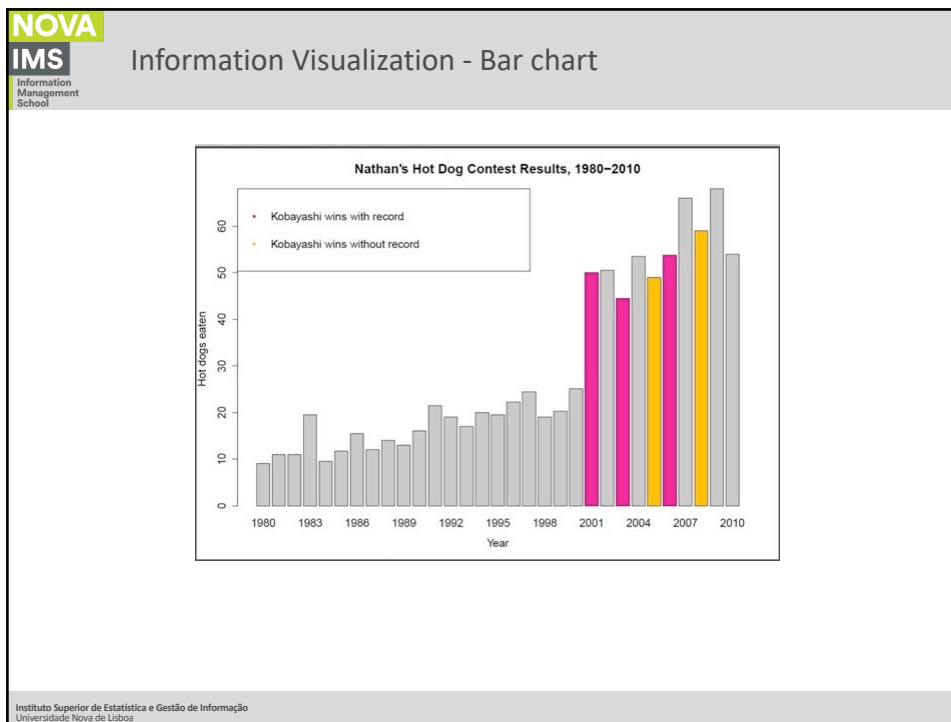
44



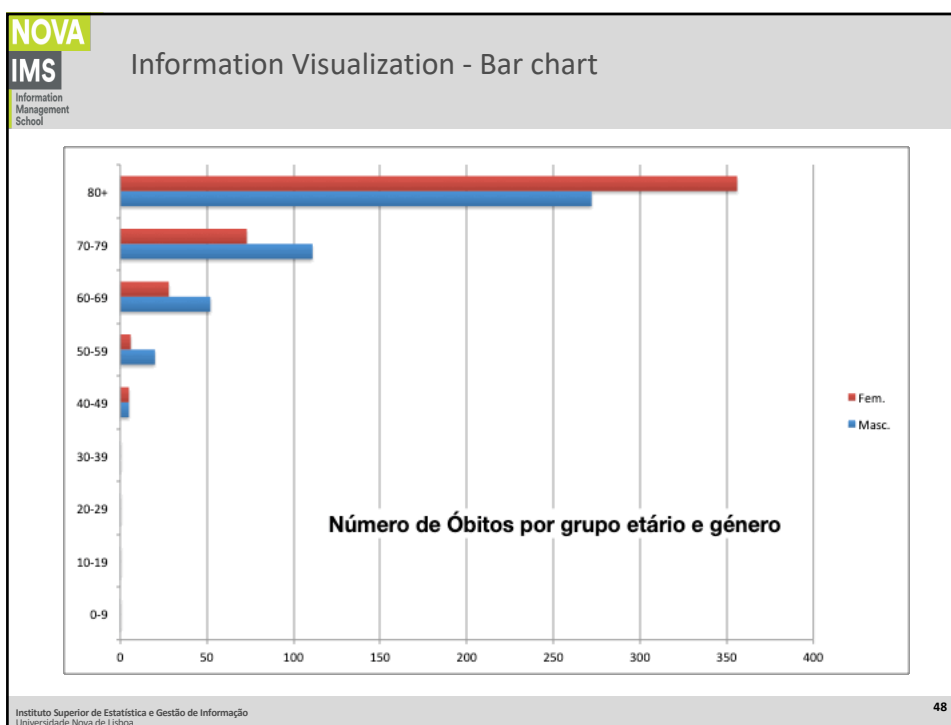
45



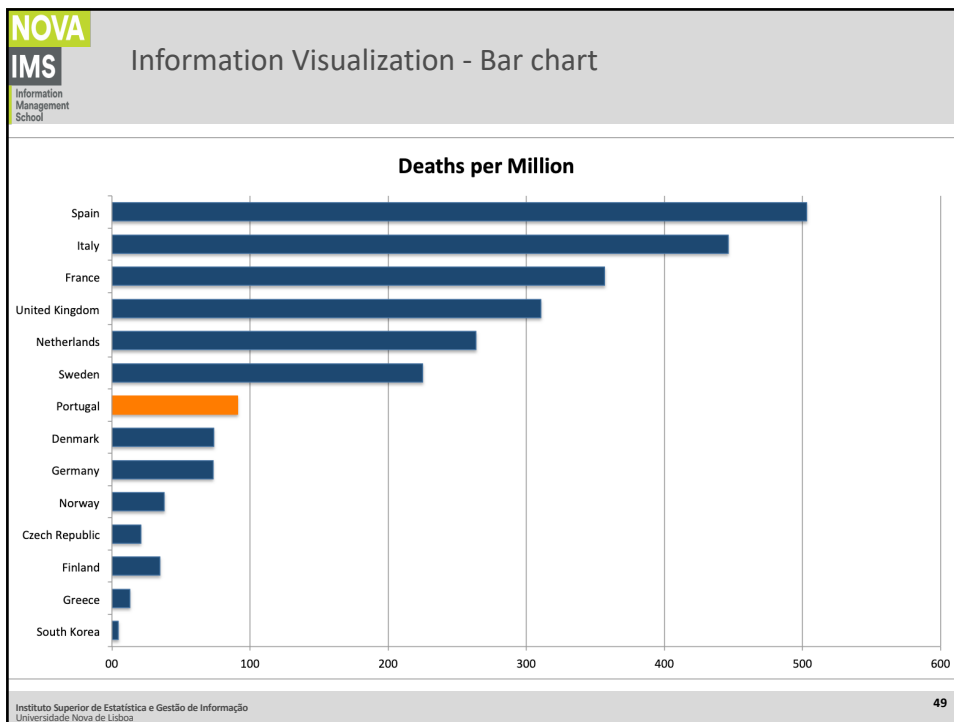
46



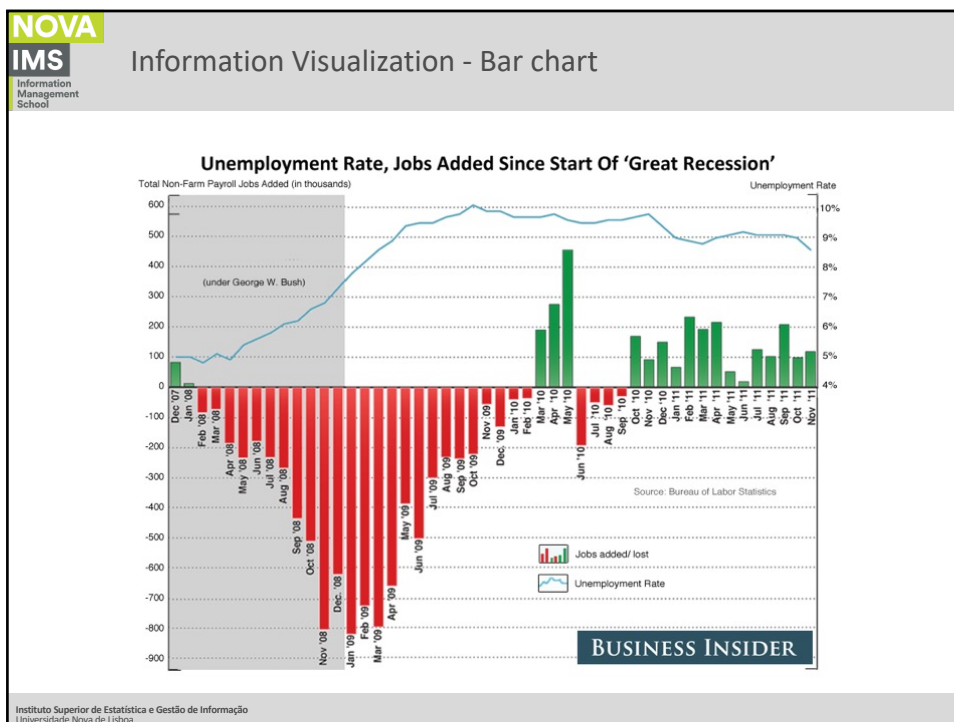
47



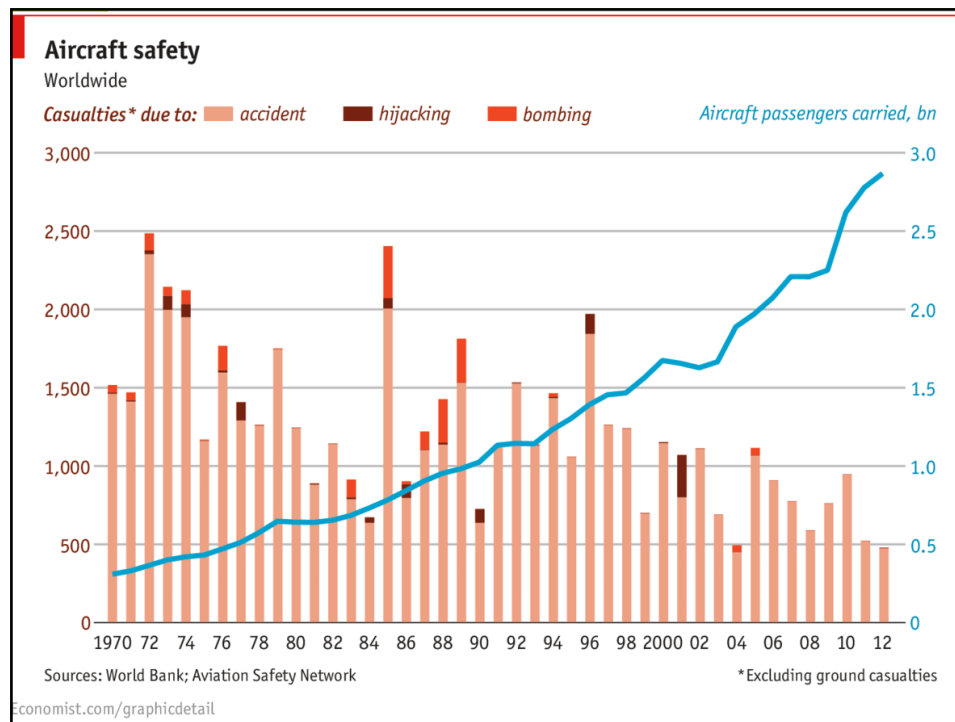
48



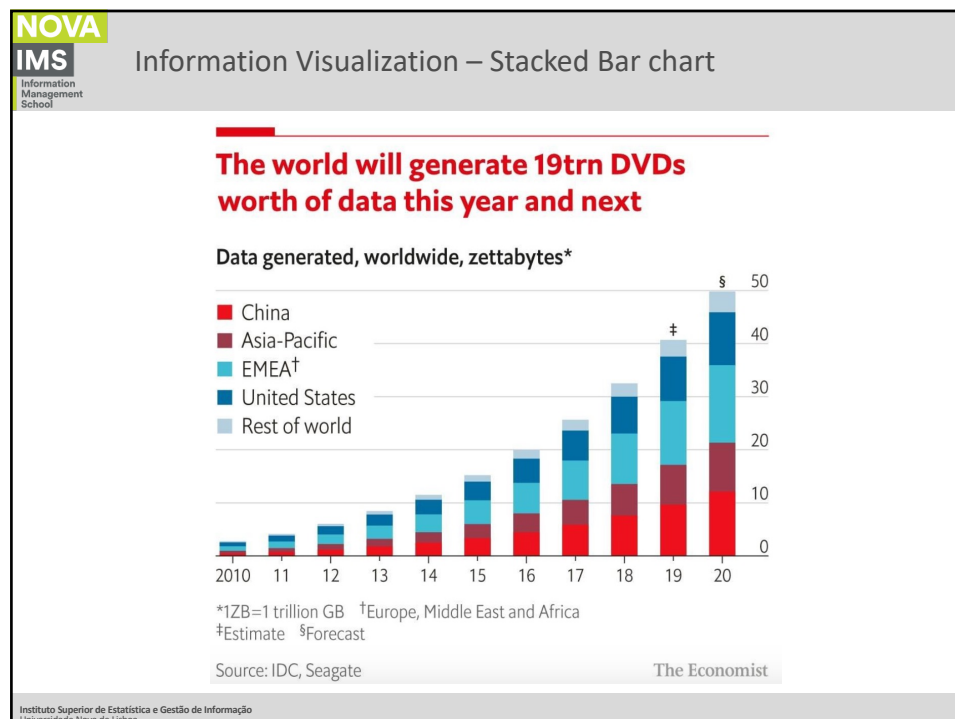
49



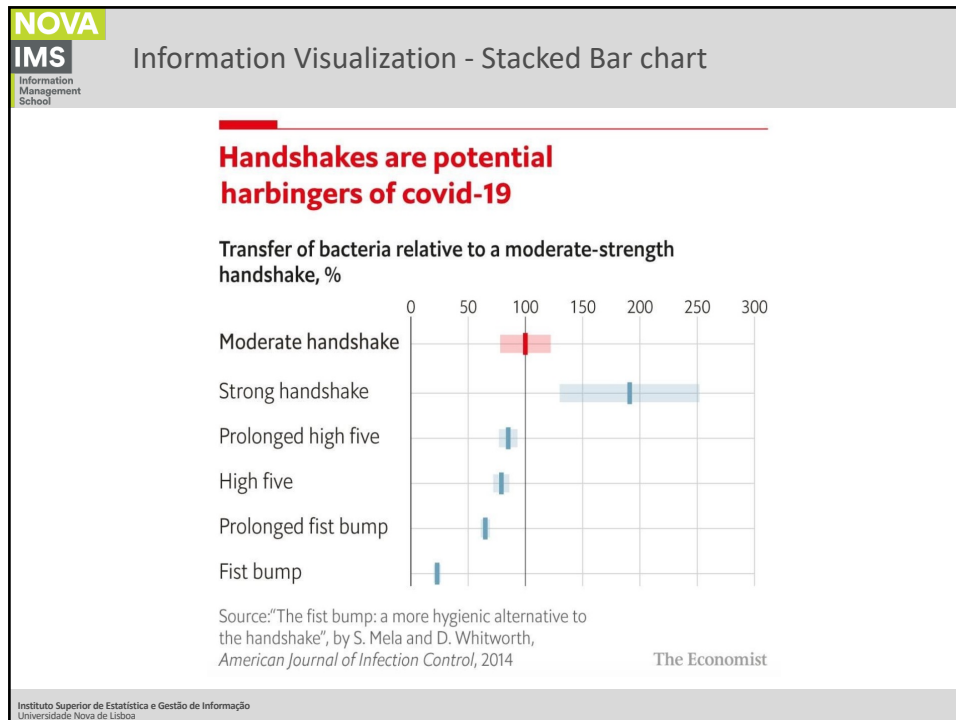
50



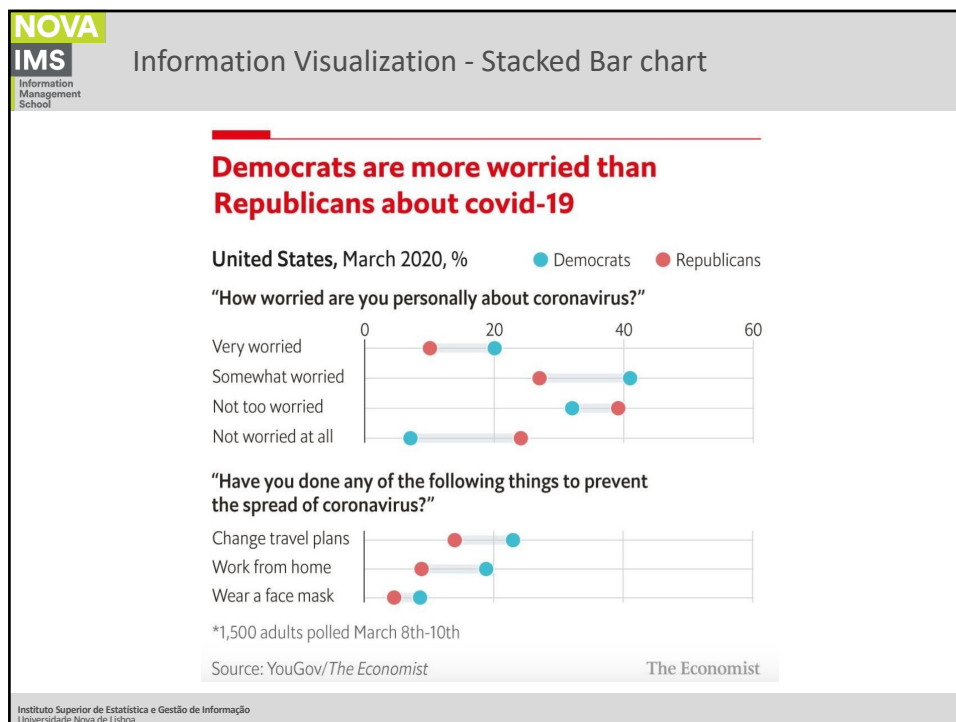
51



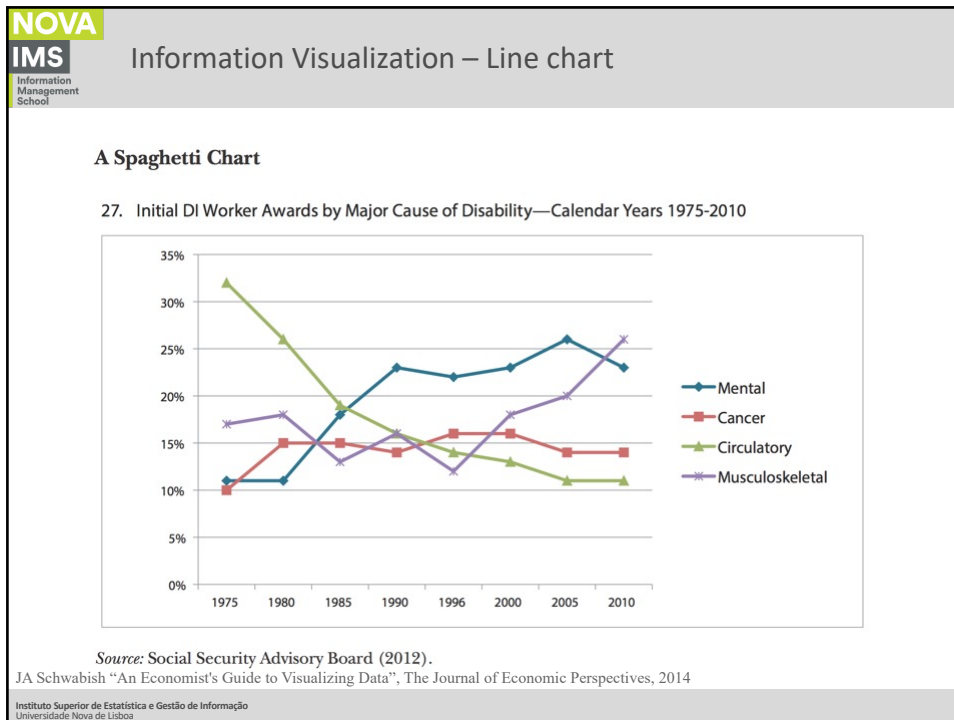
52



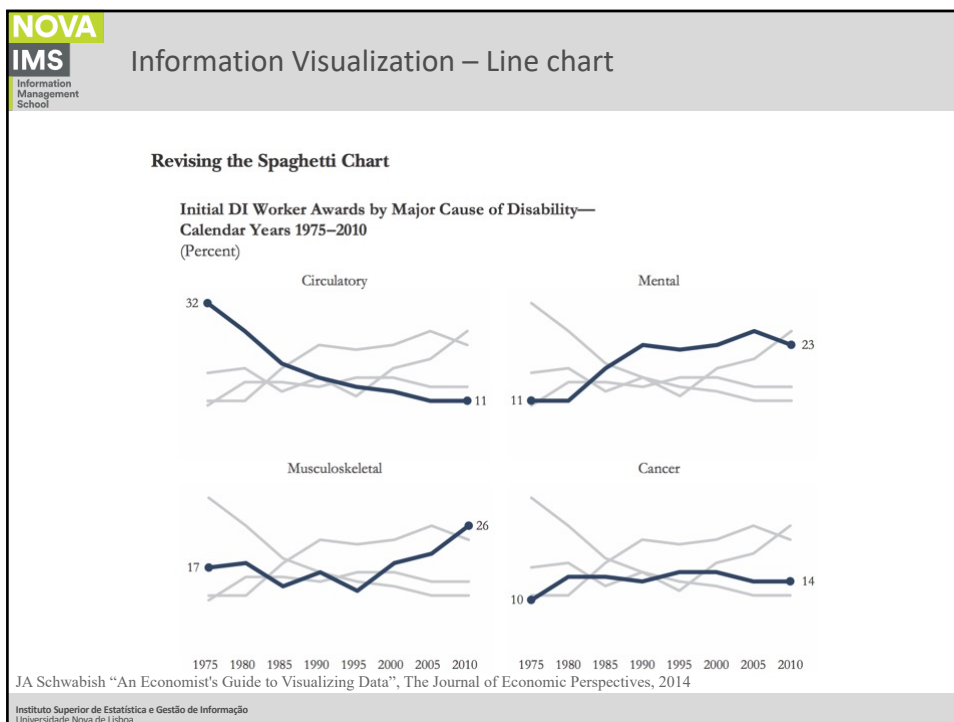
53



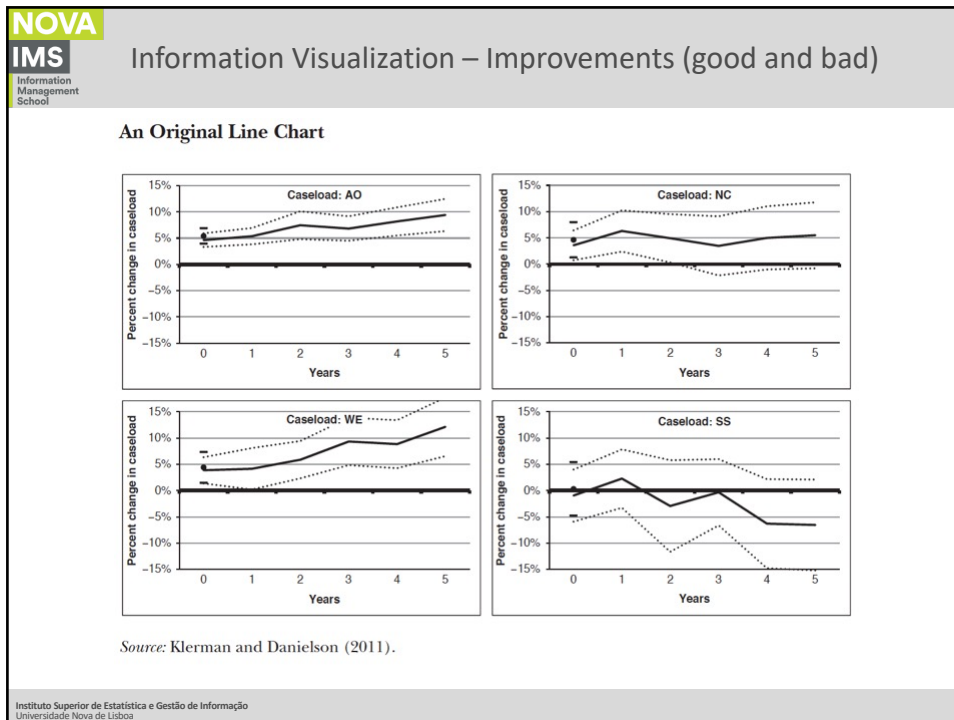
54



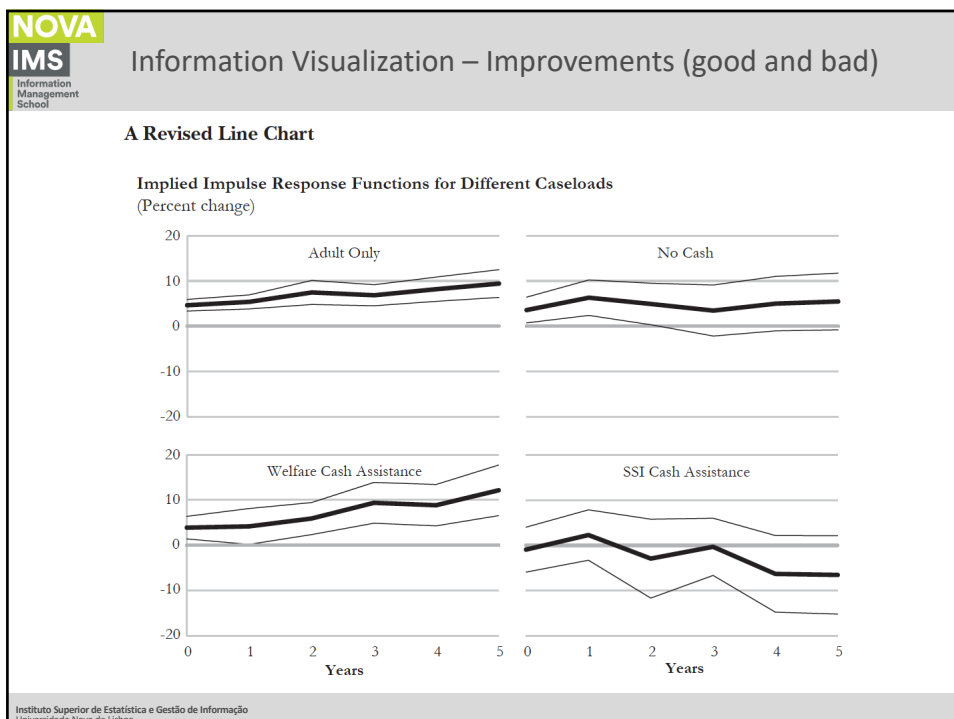
55



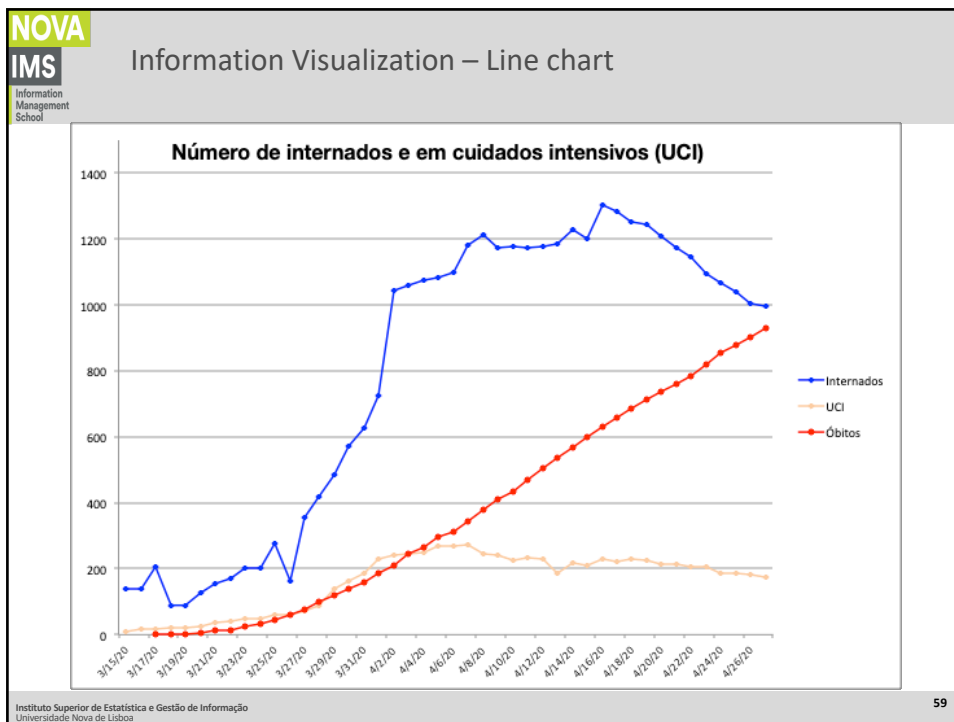
56



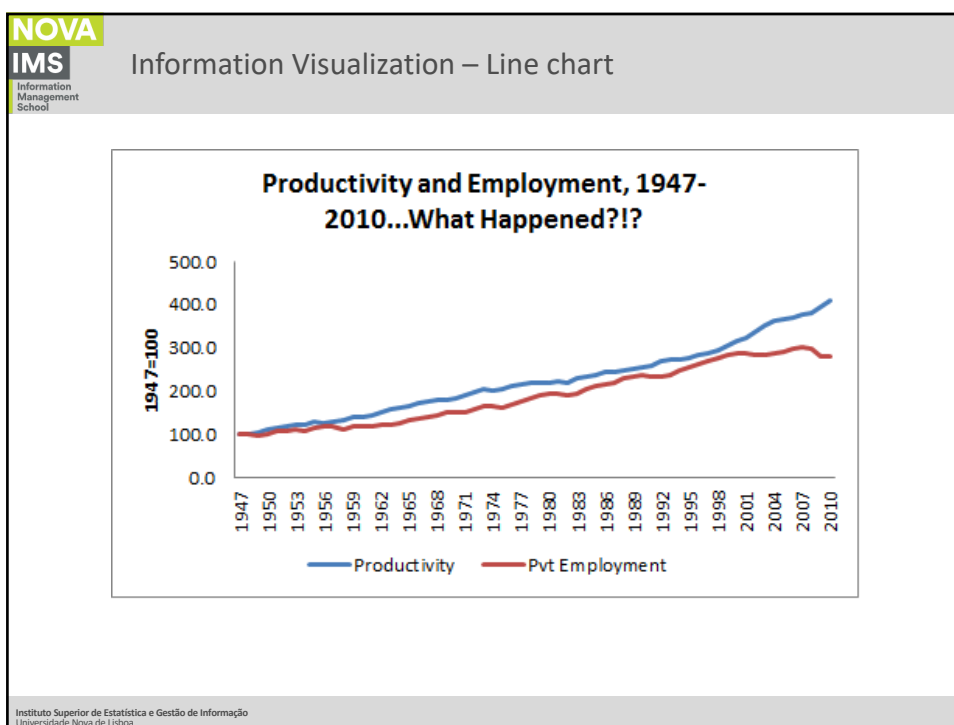
57



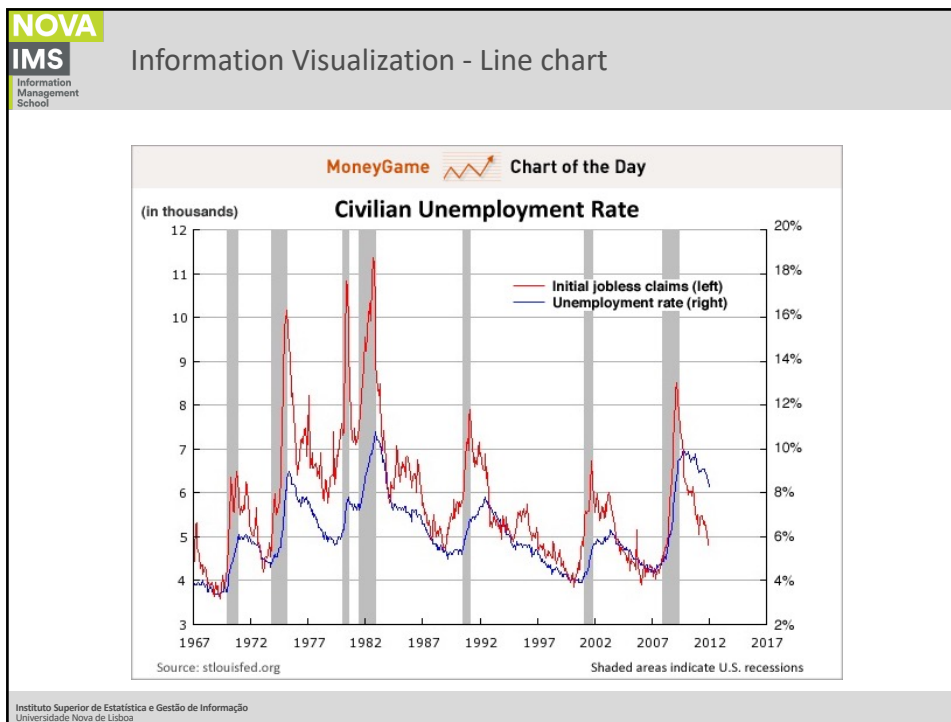
58



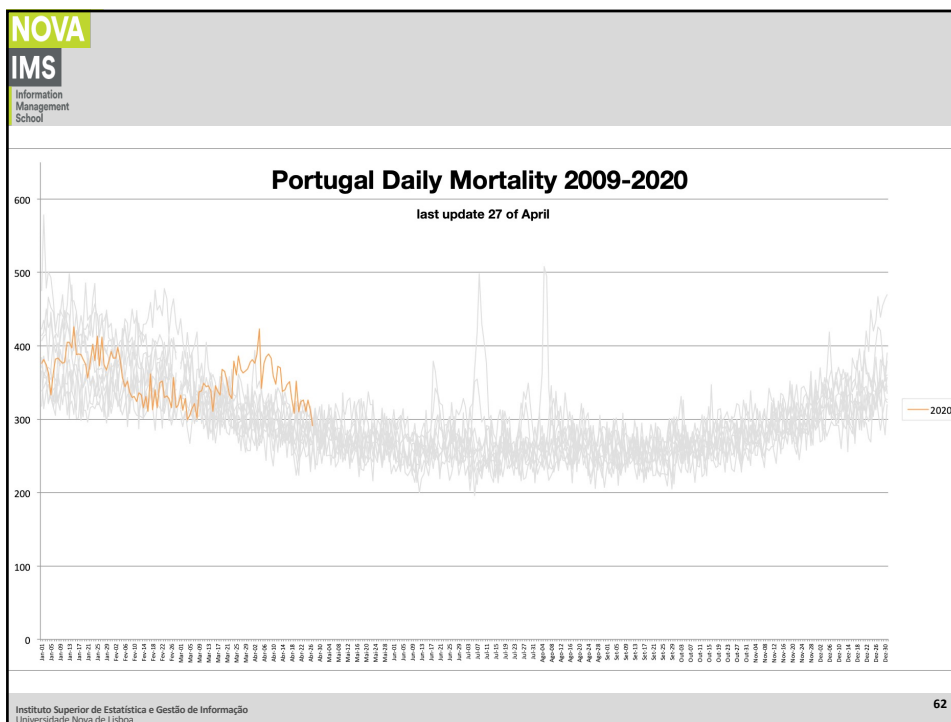
59



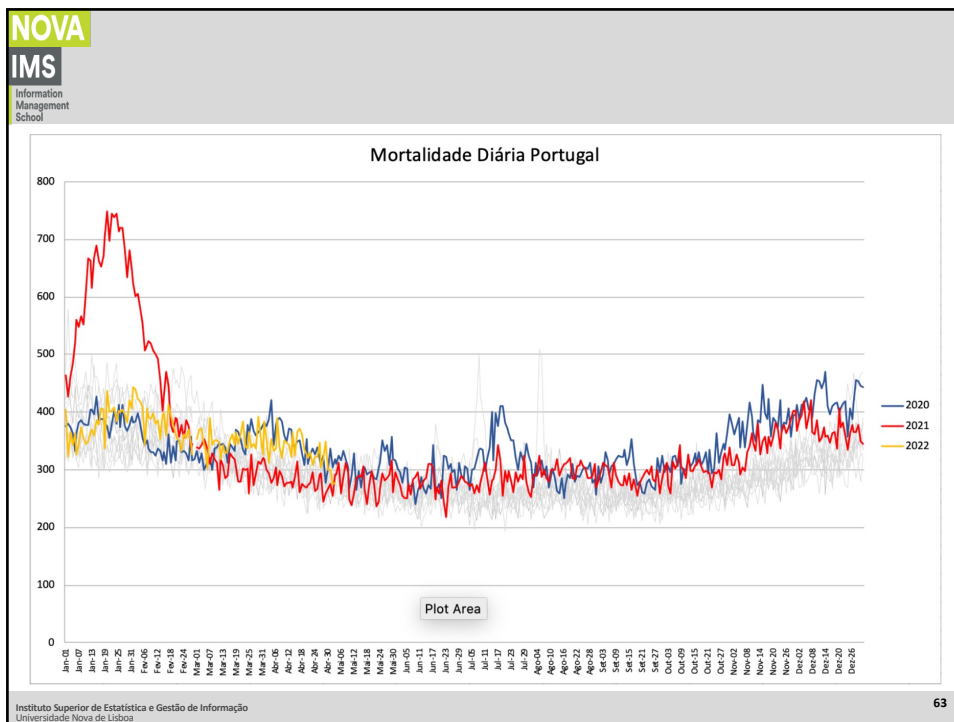
60



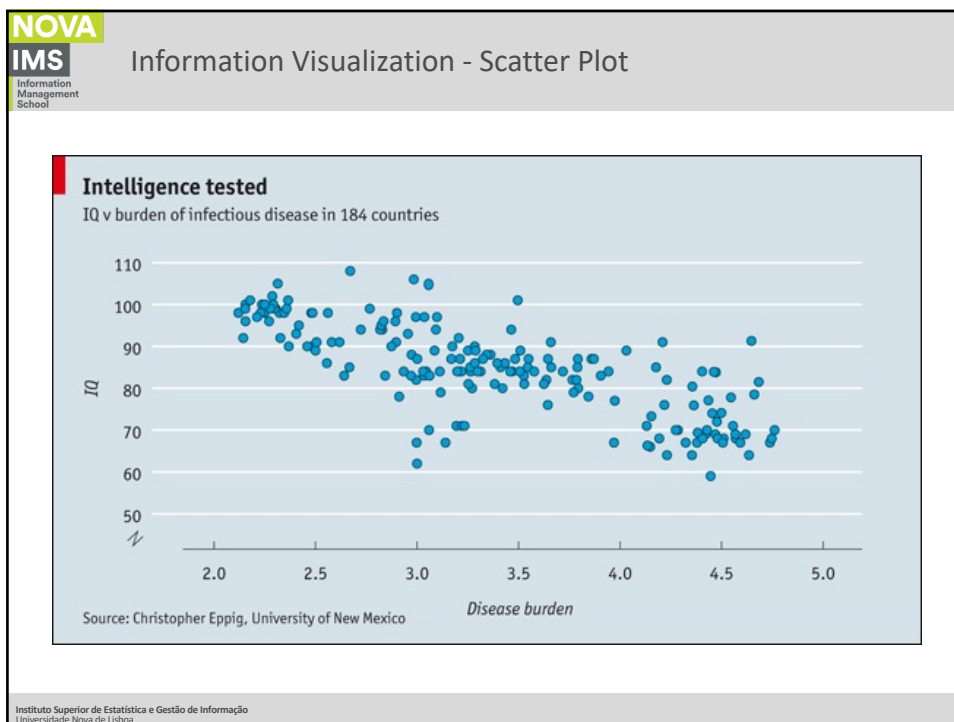
61

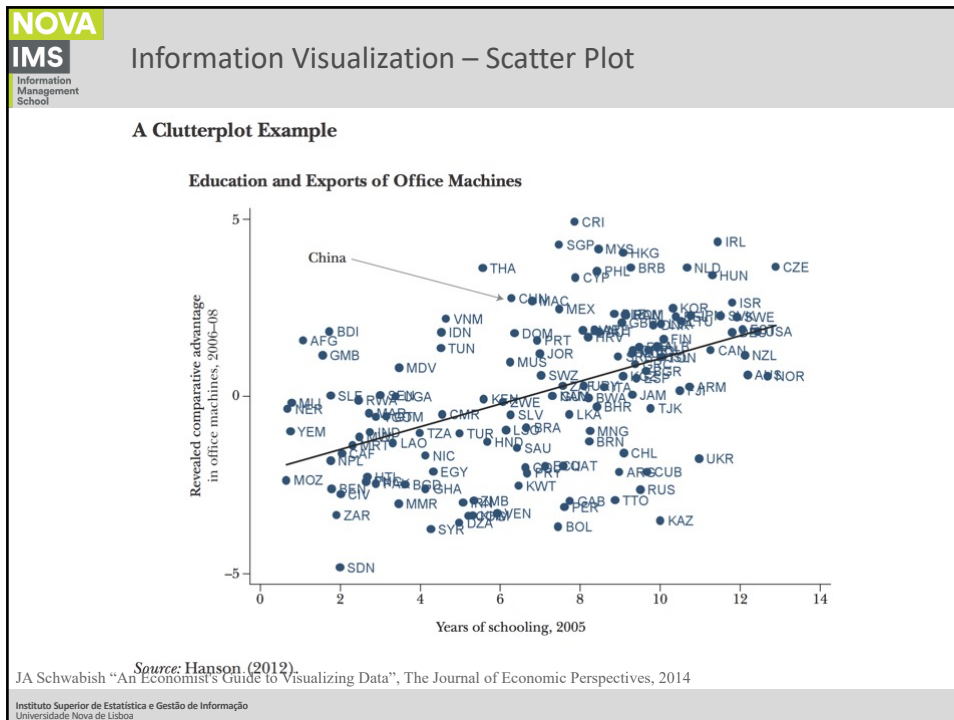


62

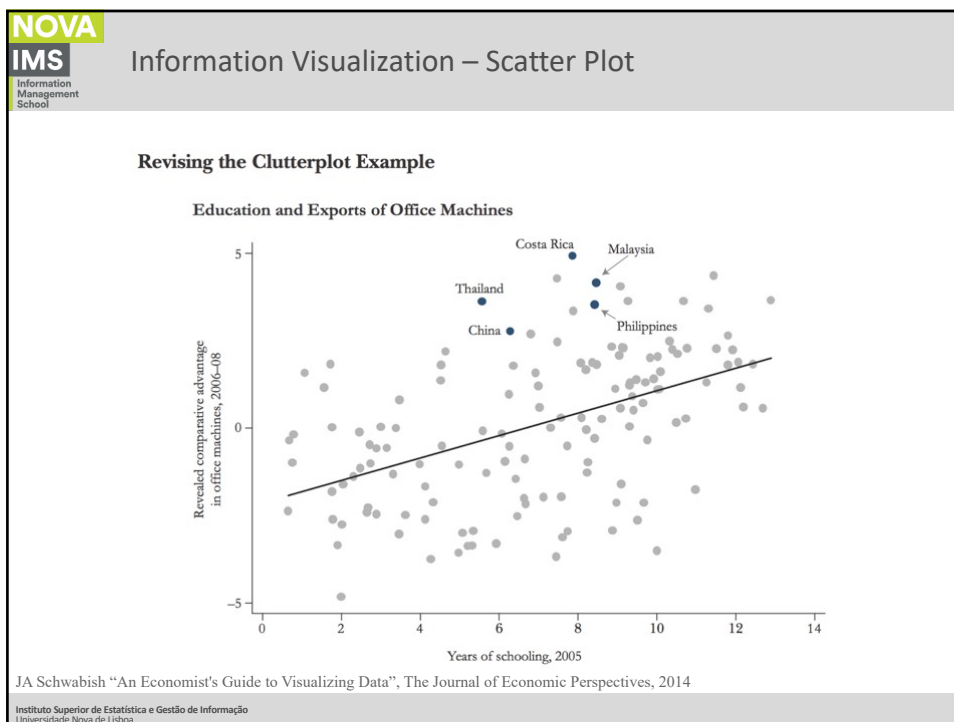


63

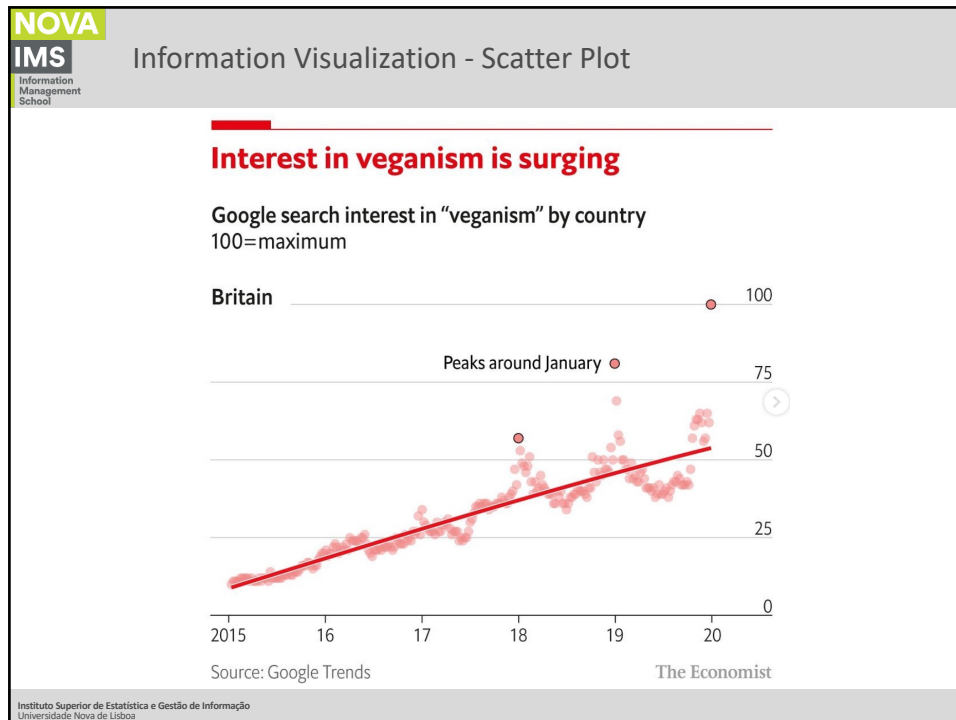




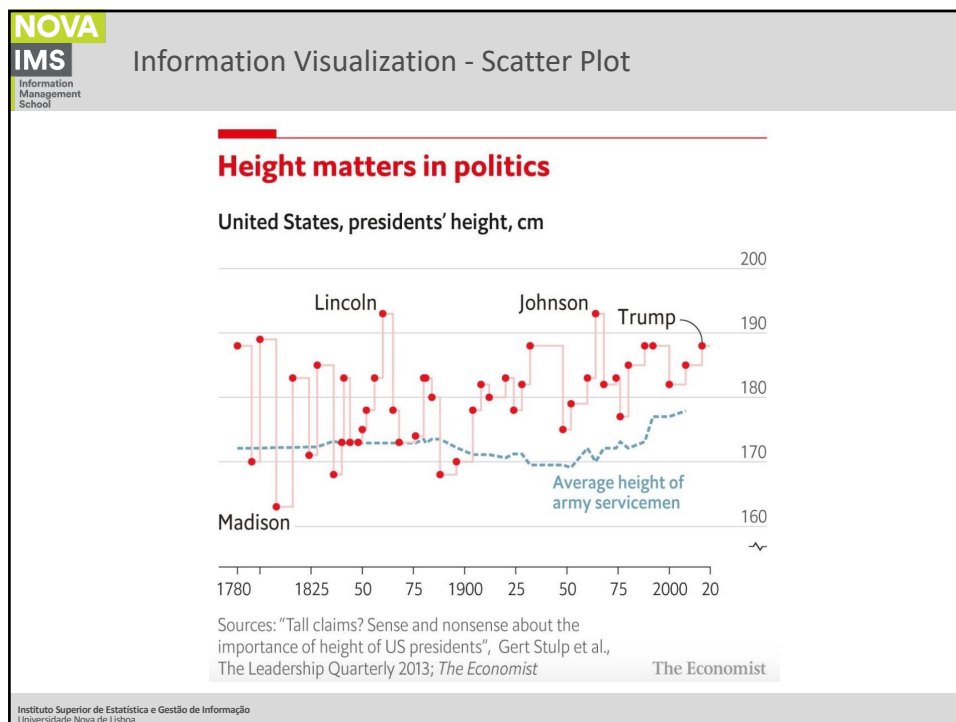
65



66



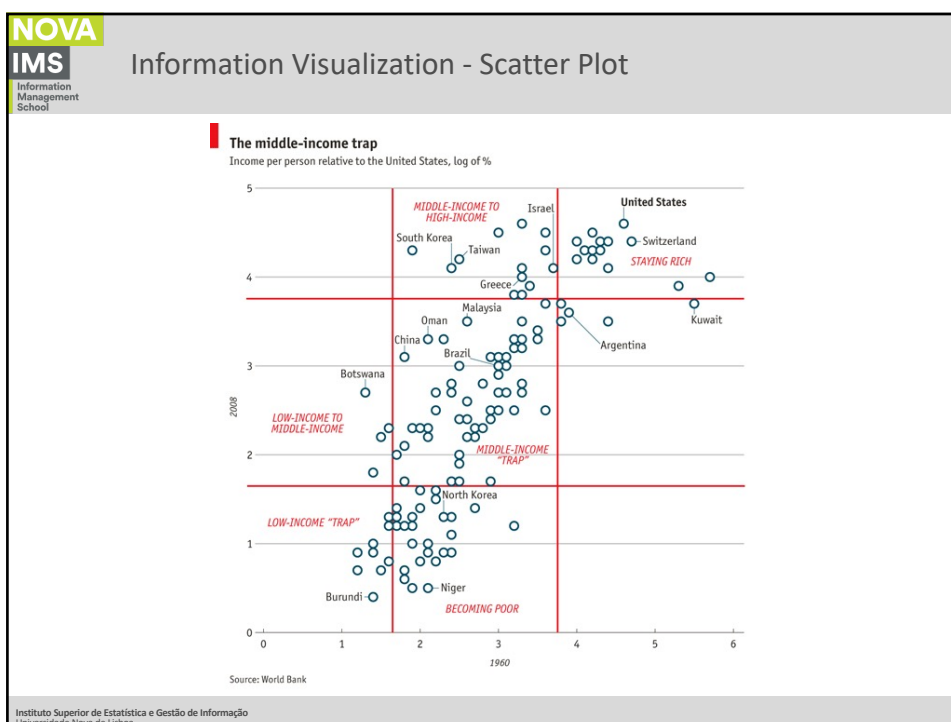
67



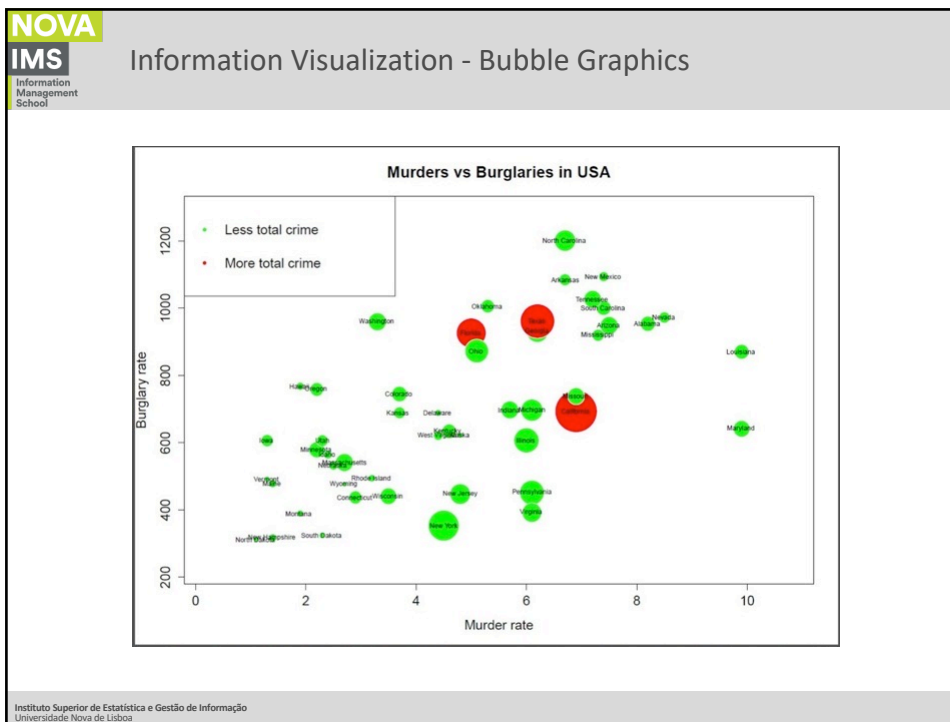
68



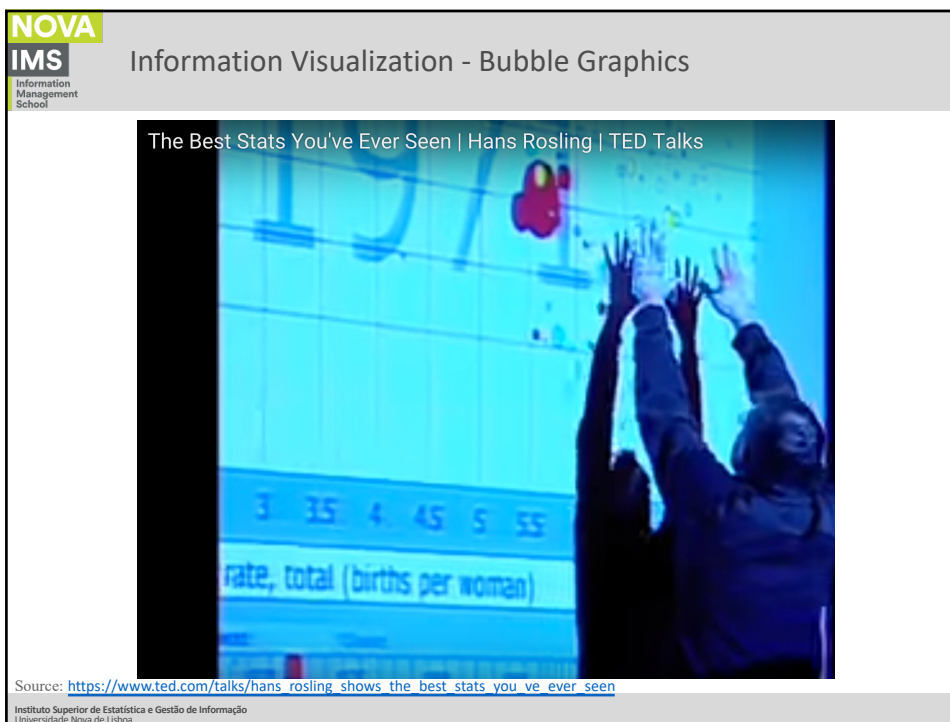
69



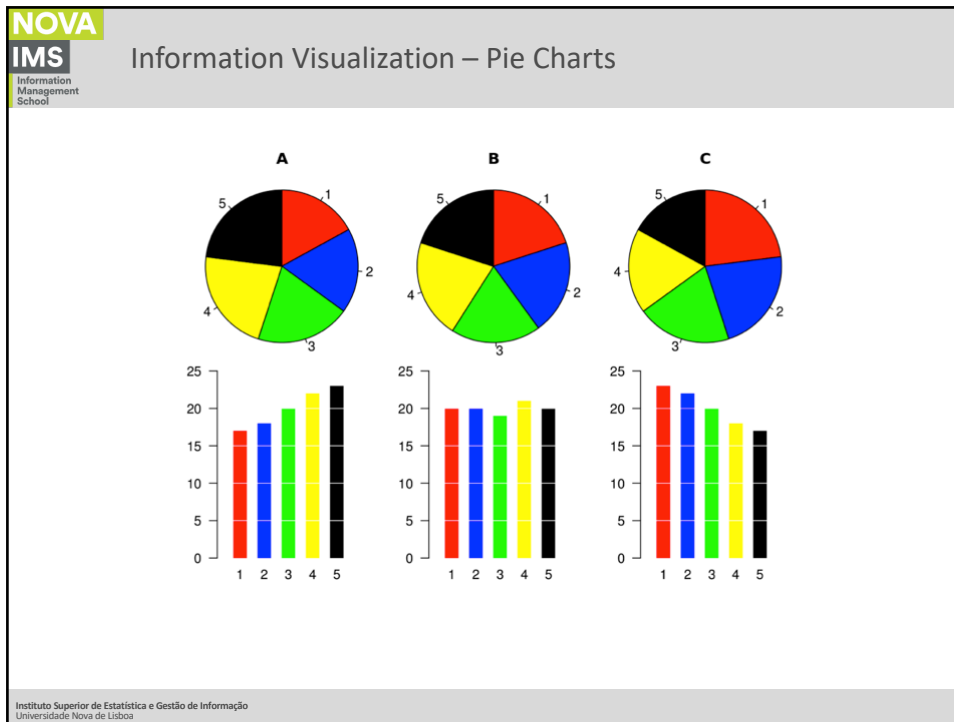
70



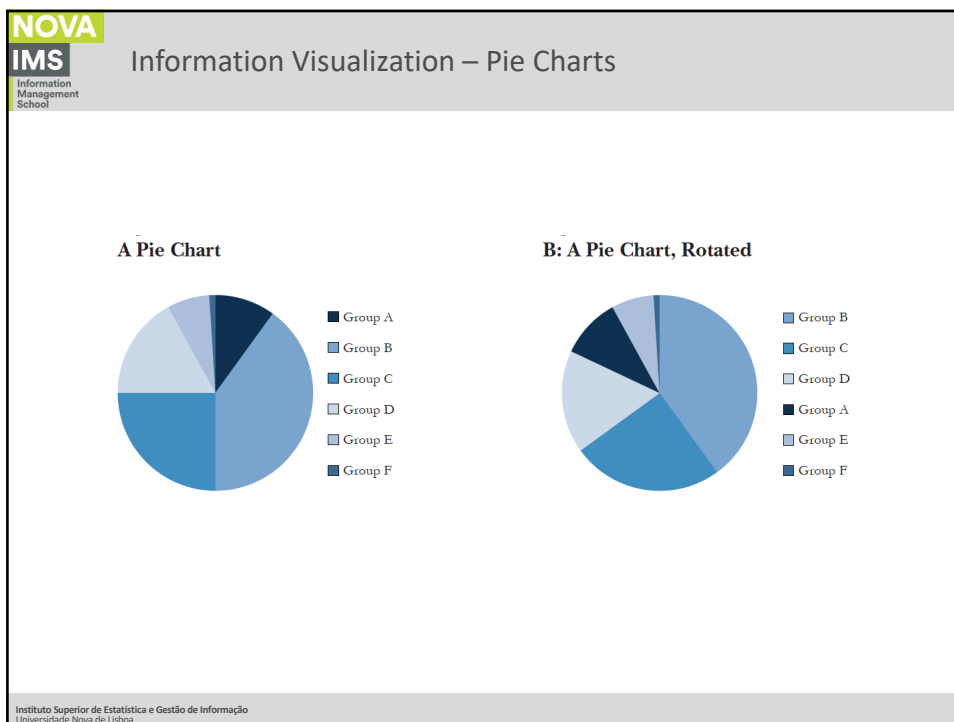
71



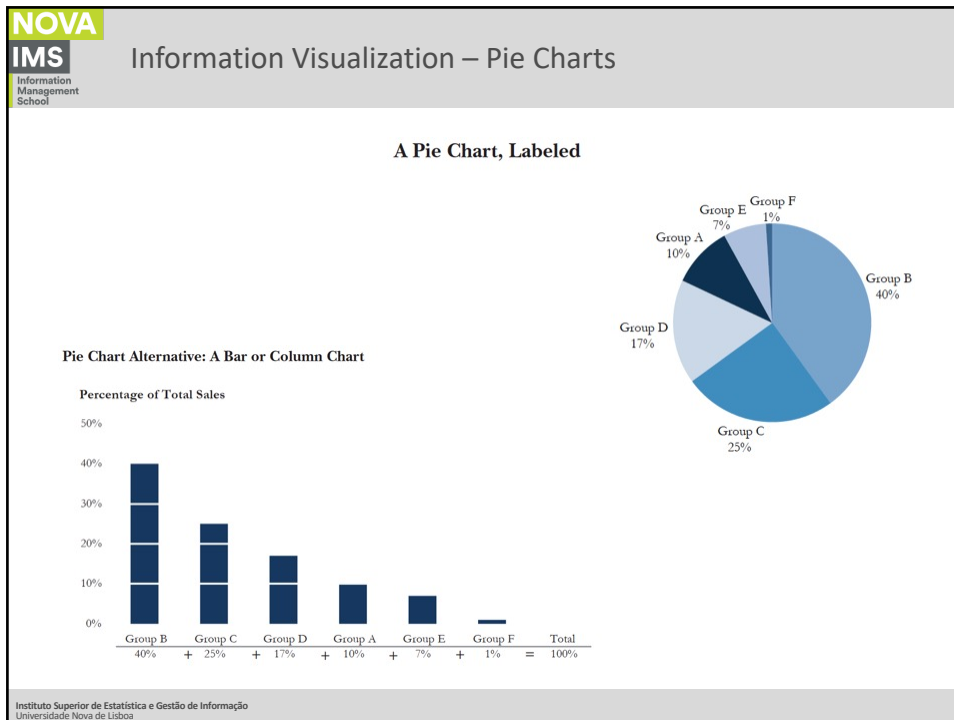
72



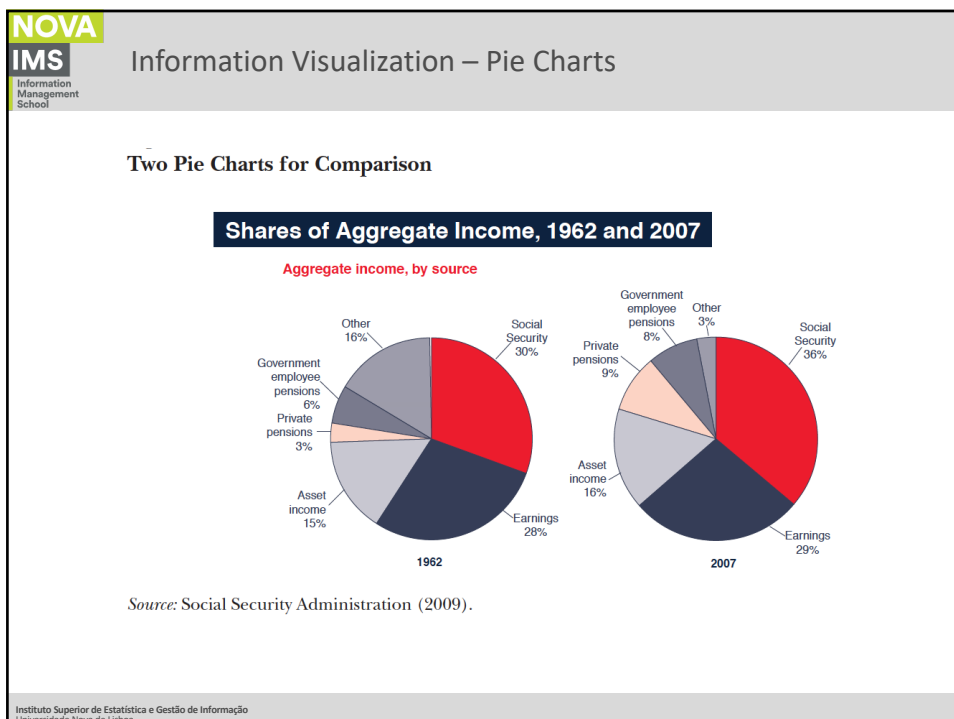
73



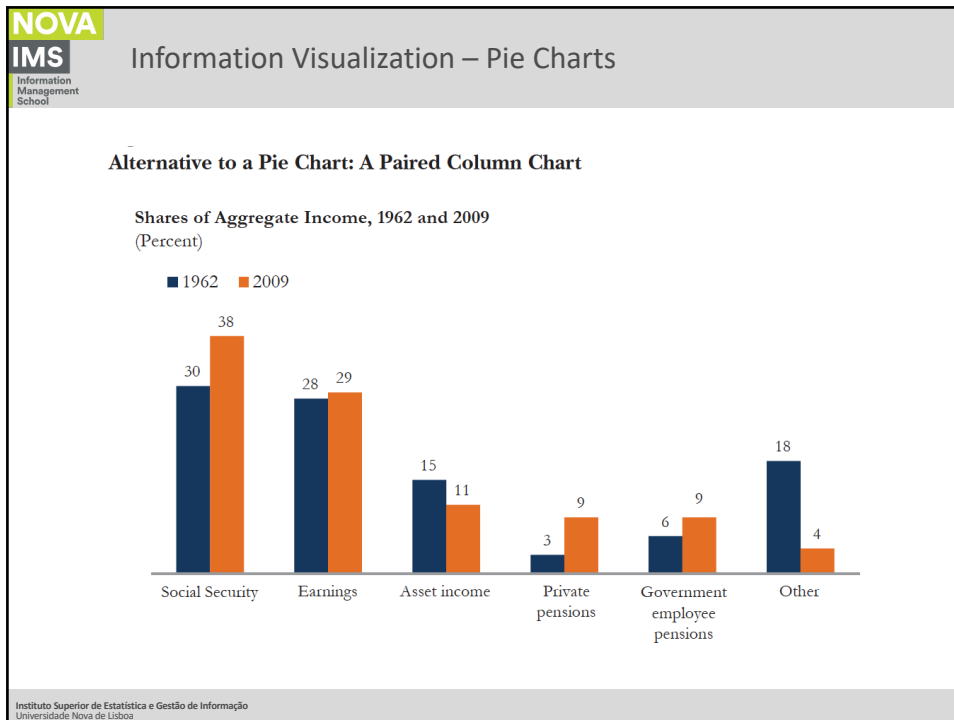
74



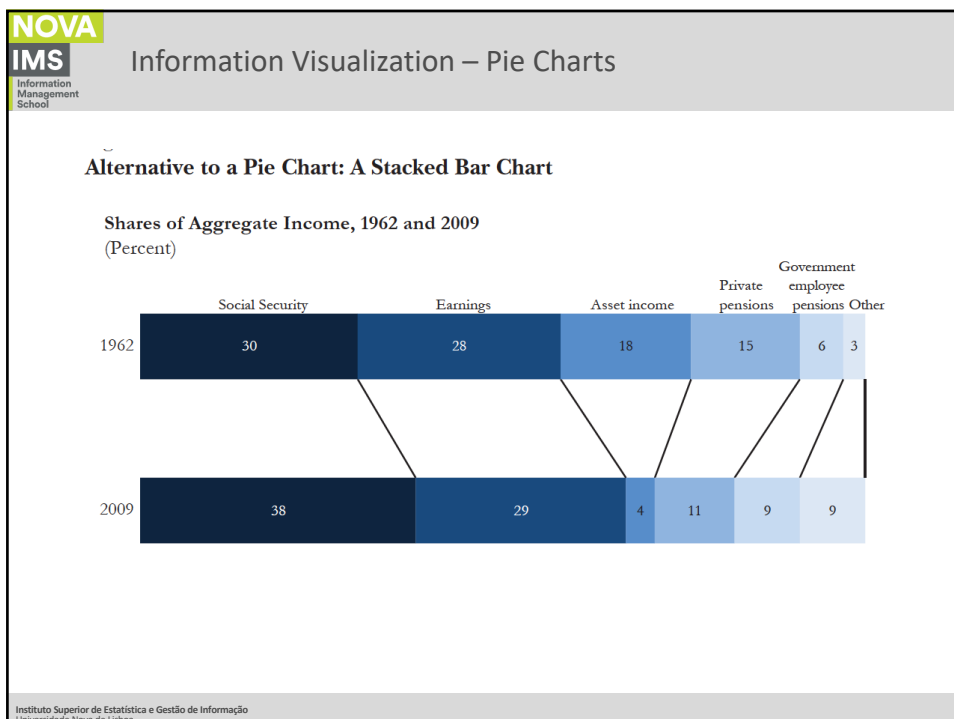
75



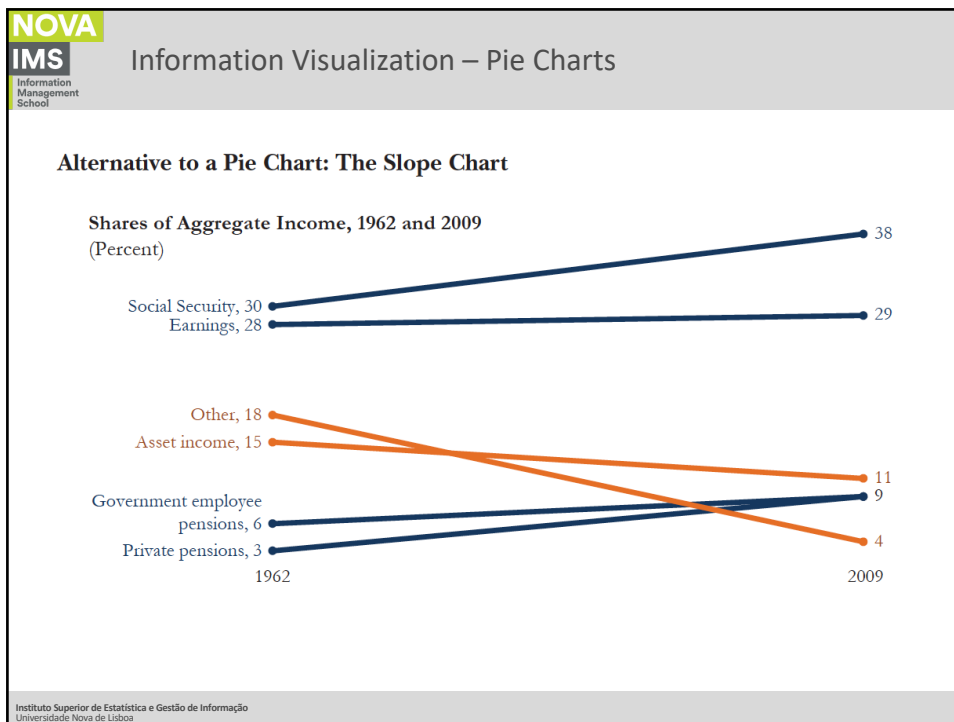
76



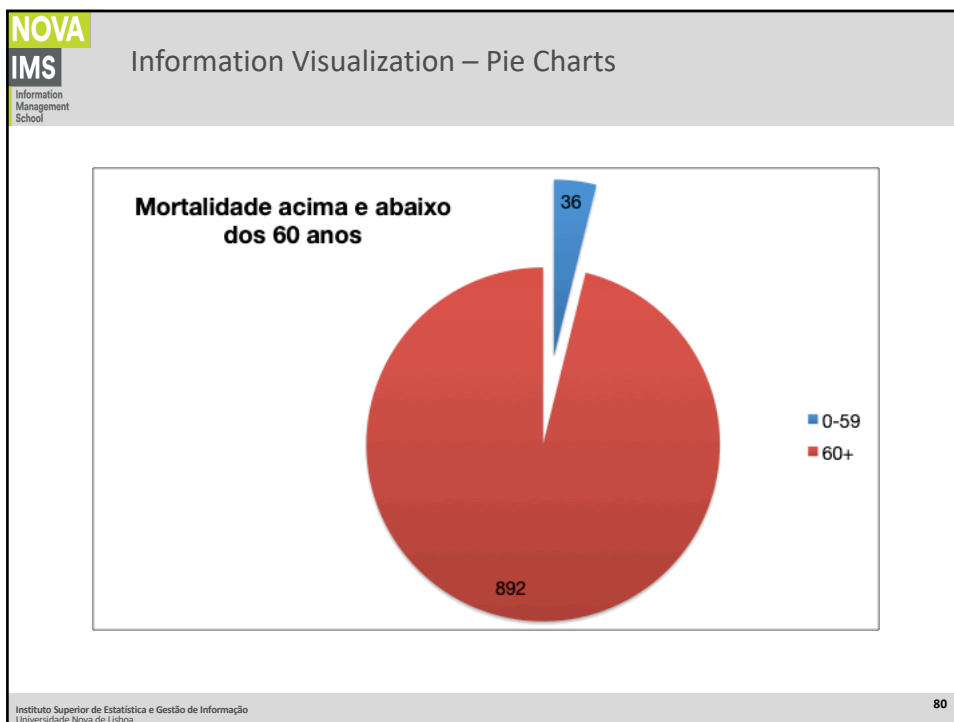
77



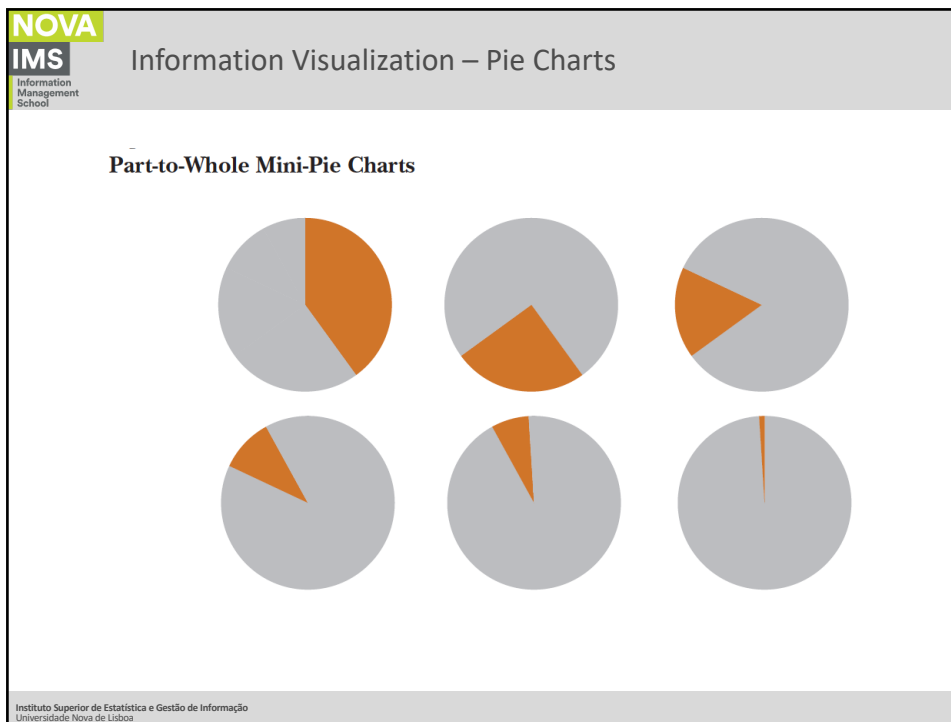
78



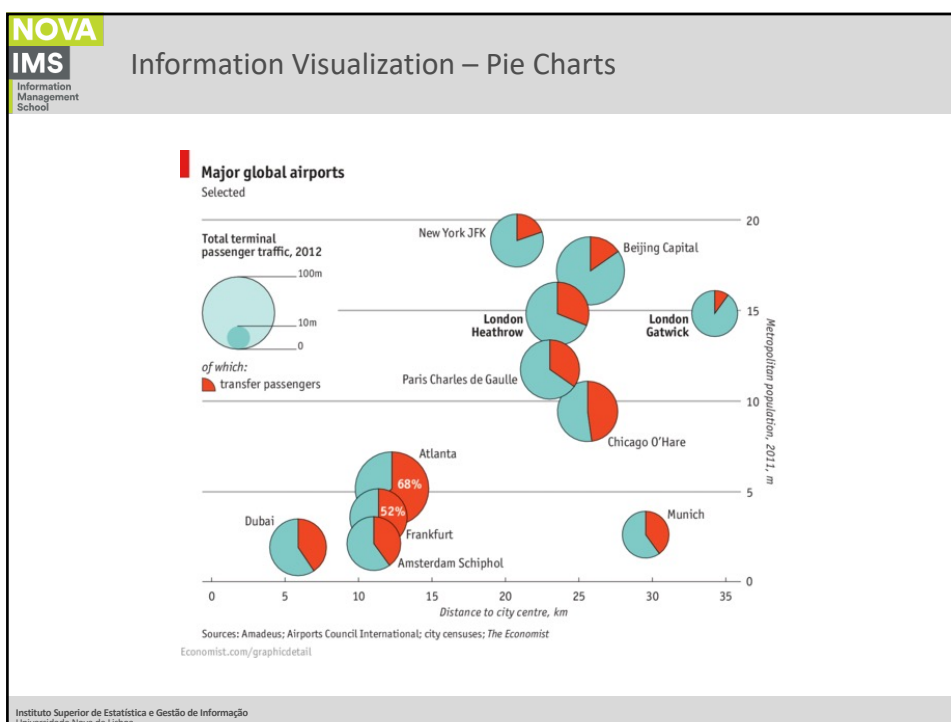
79



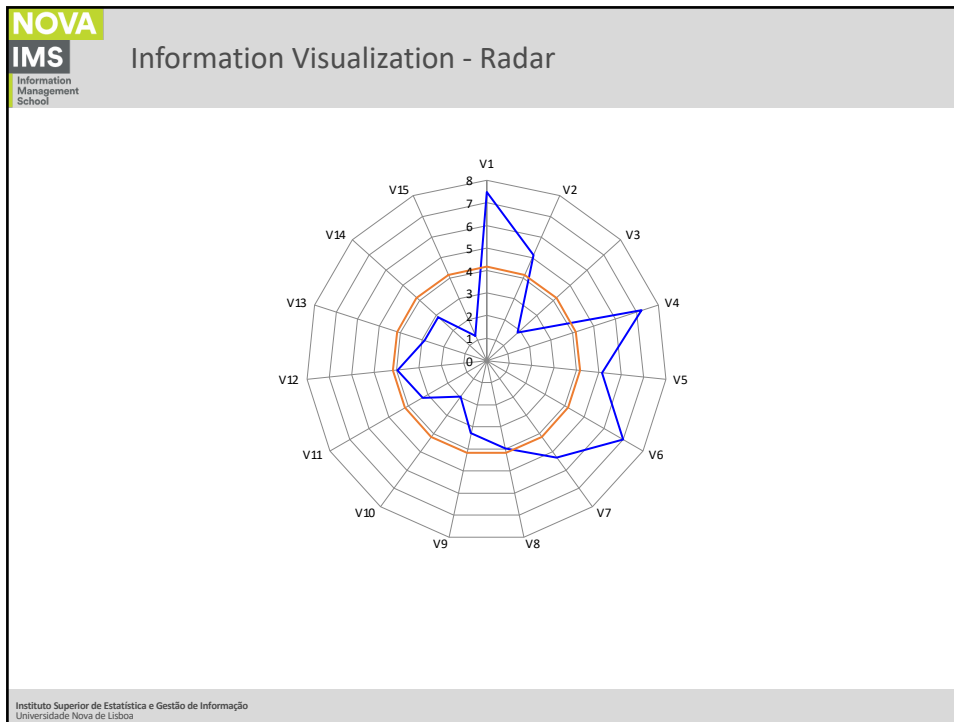
80



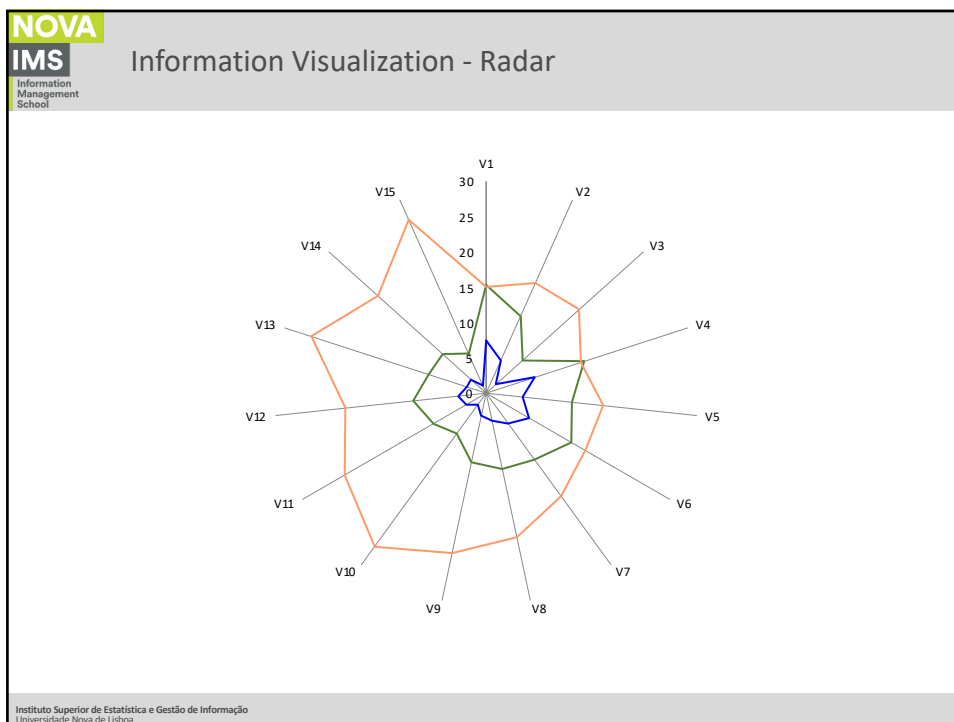
81



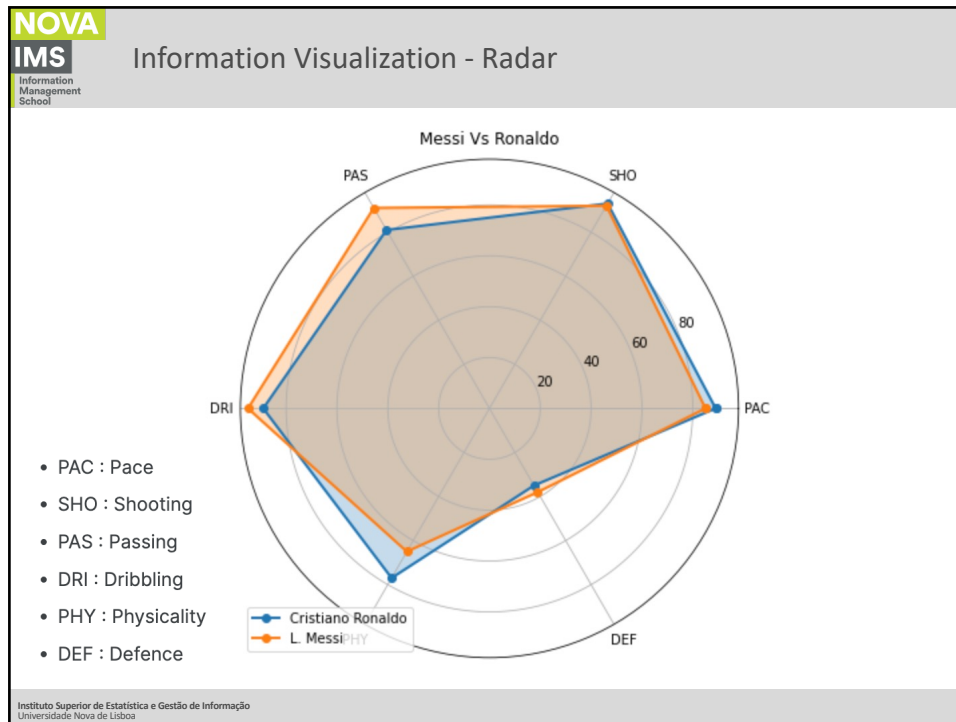
82



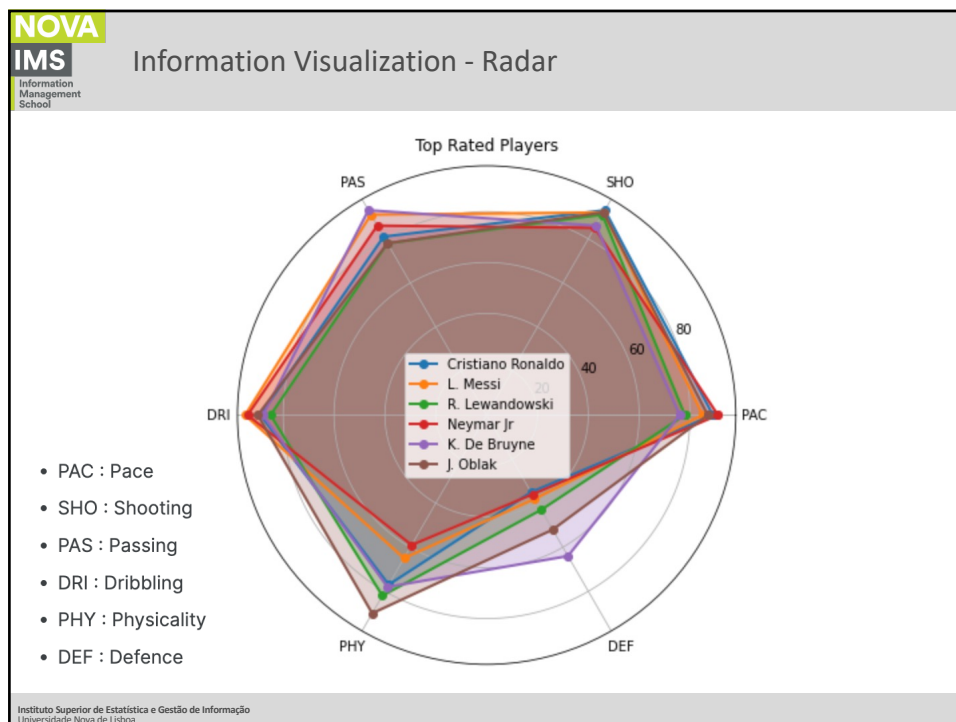
83



84



85



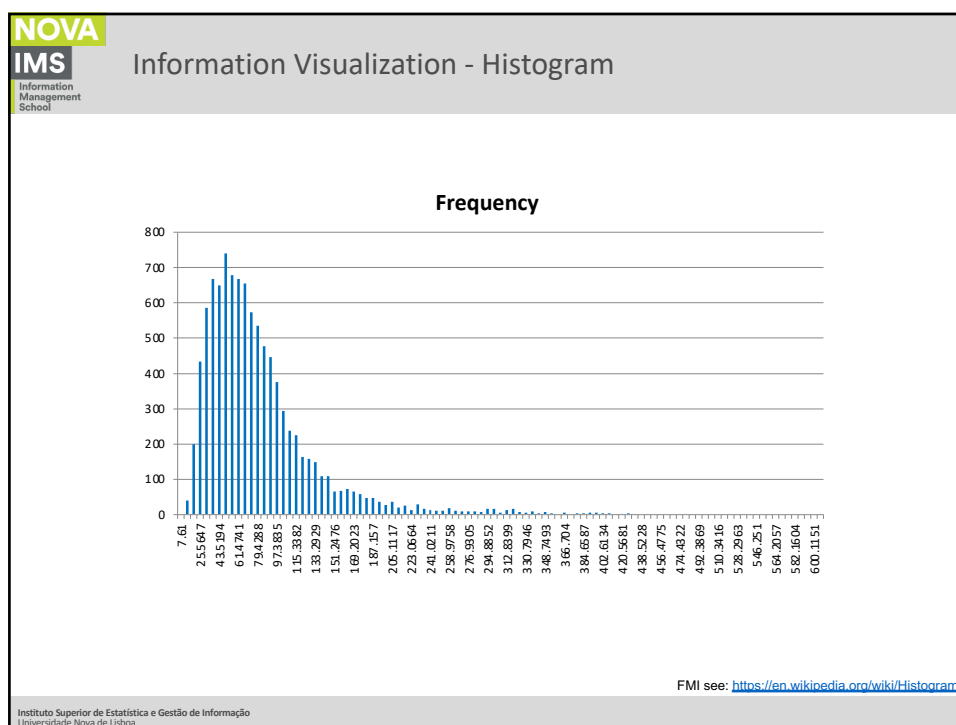
86

NOVA
IMS
Information Management School

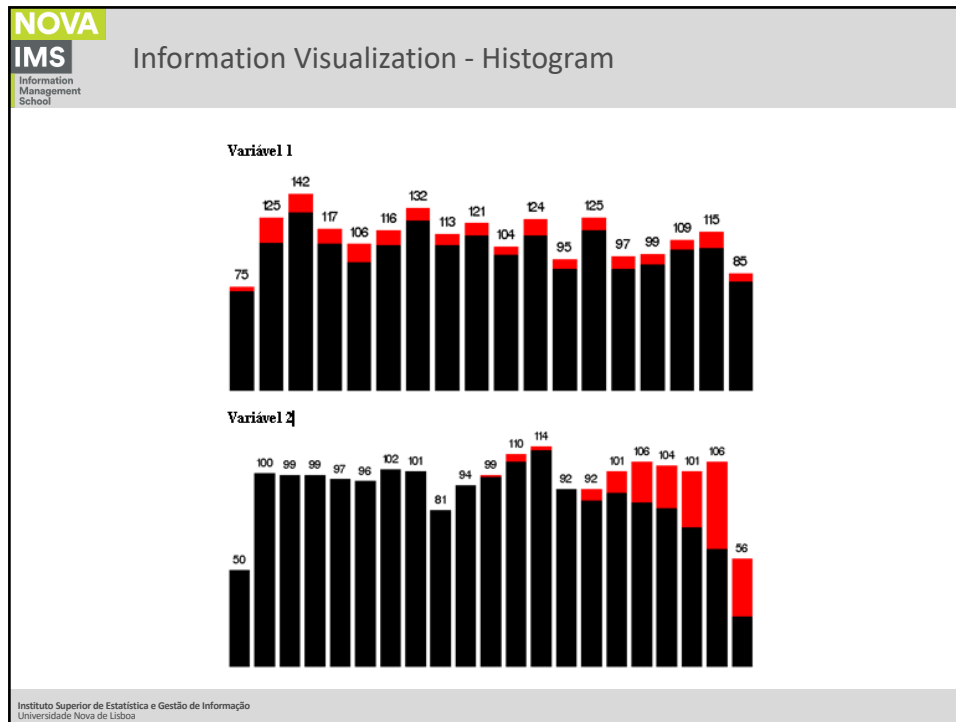
Information Visualization Analysis

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

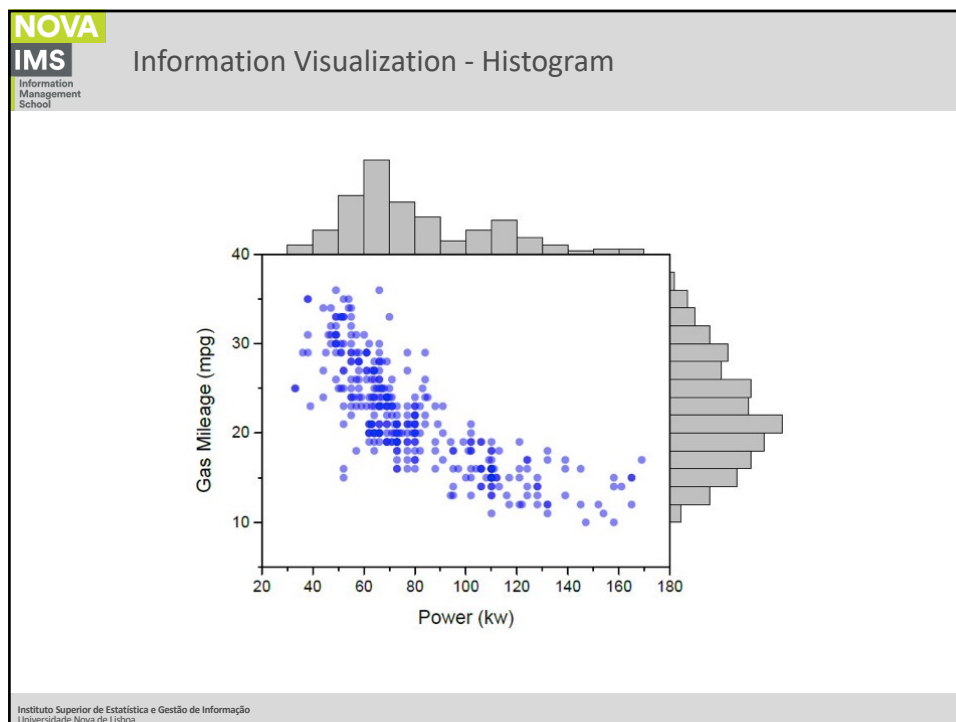
87



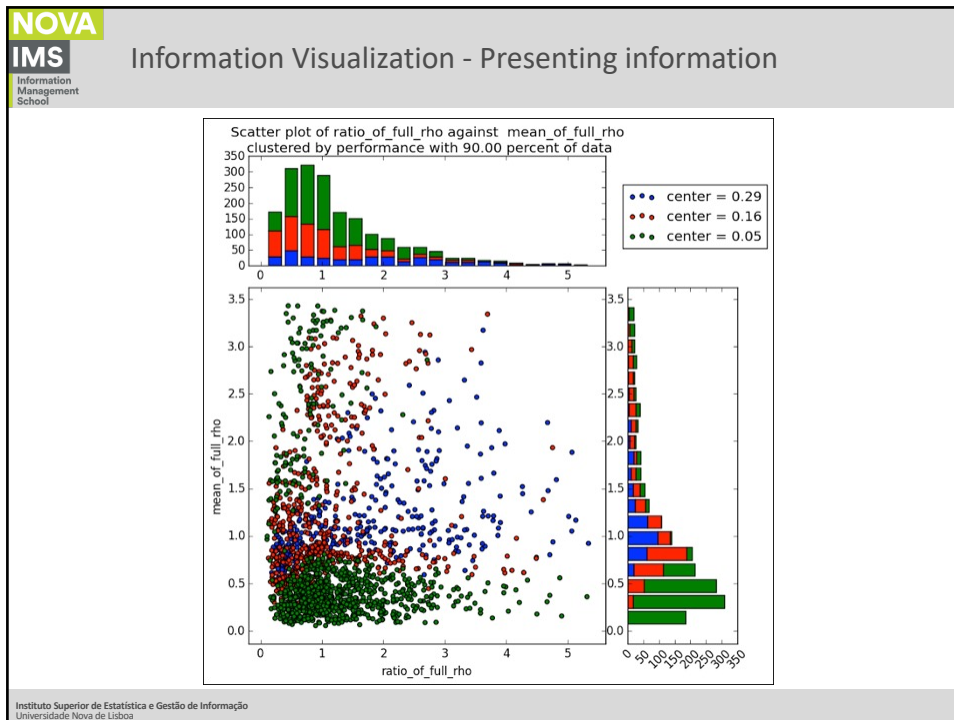
88



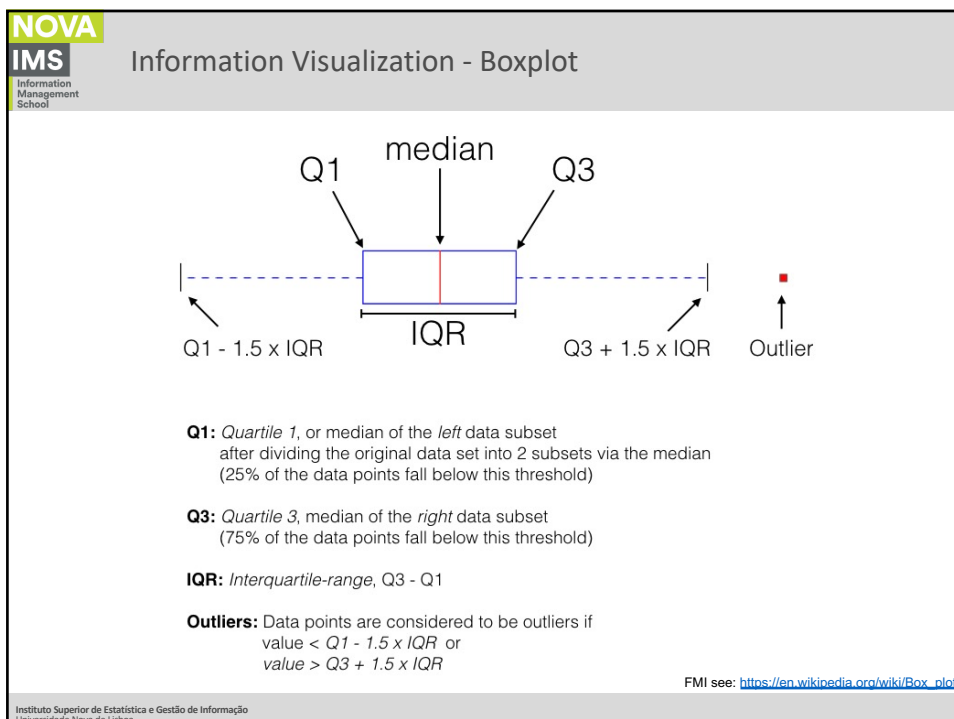
89



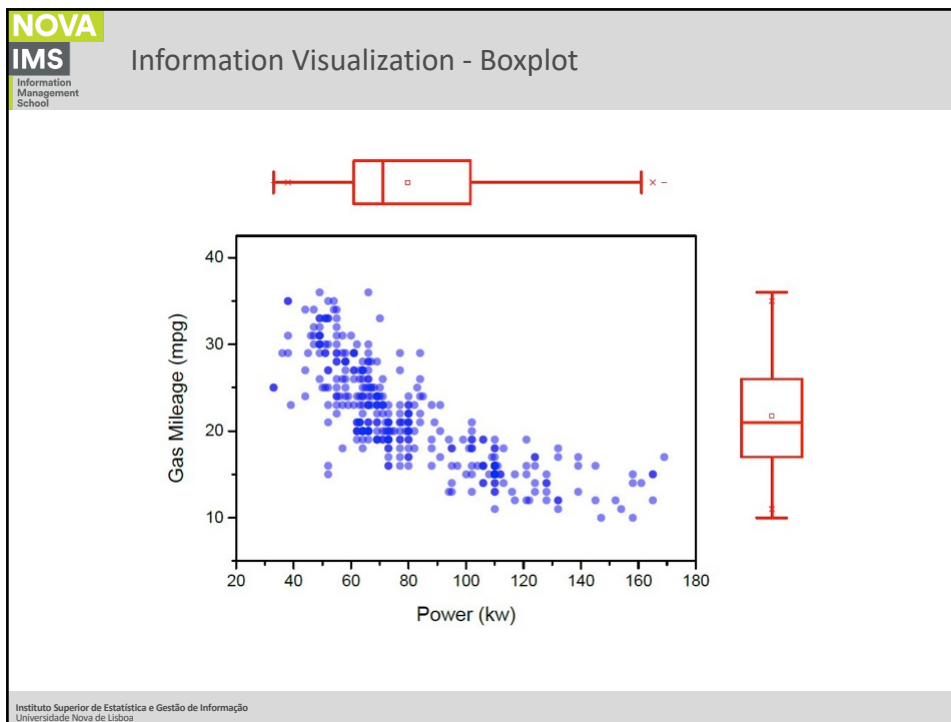
90



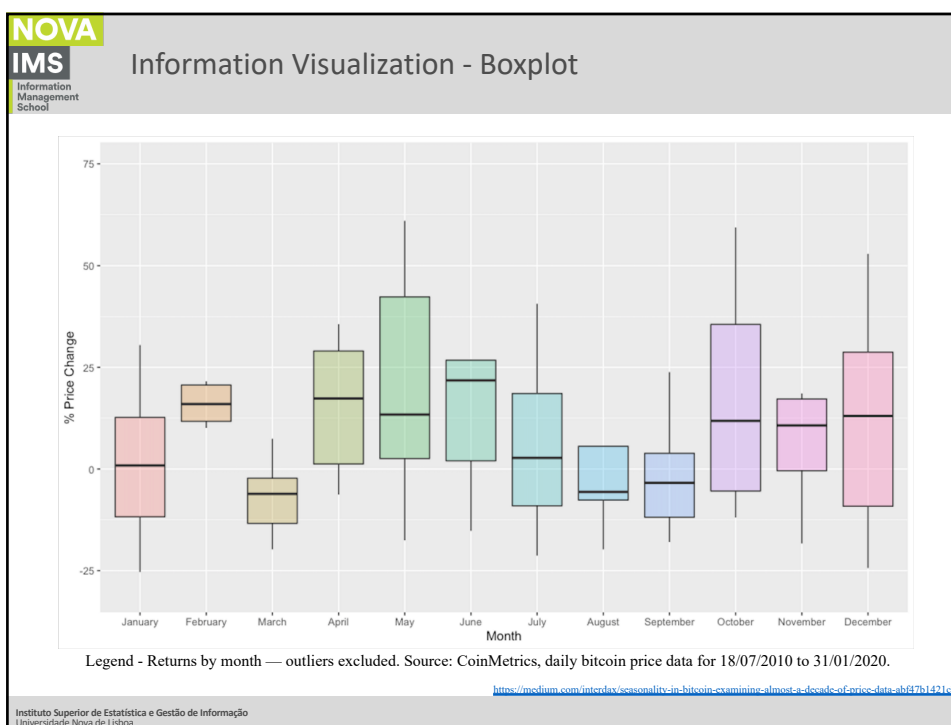
91



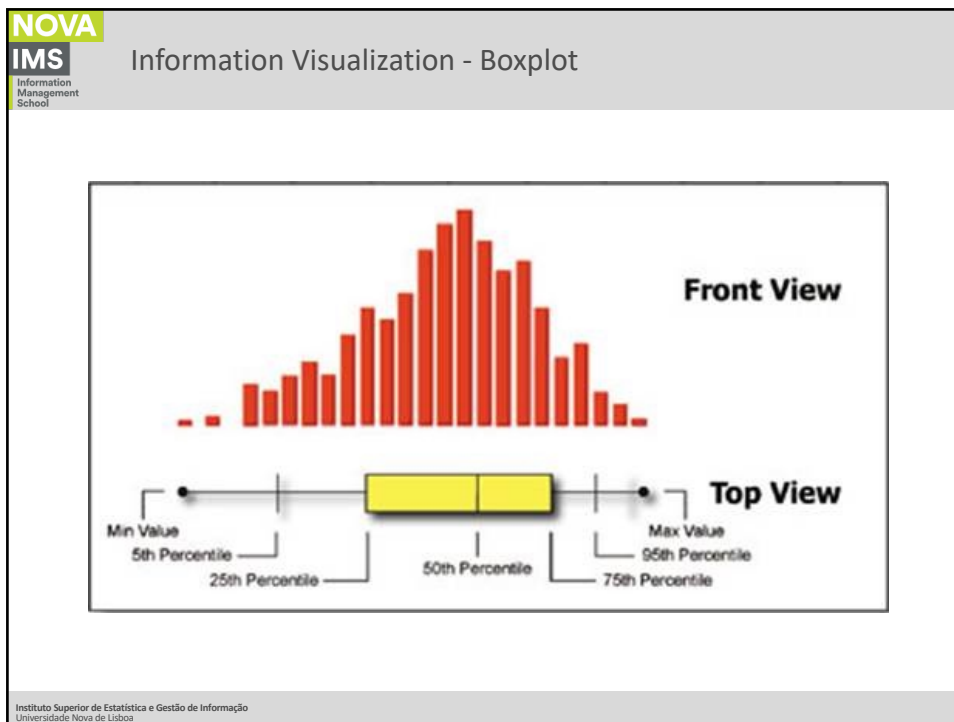
92



93



94



95

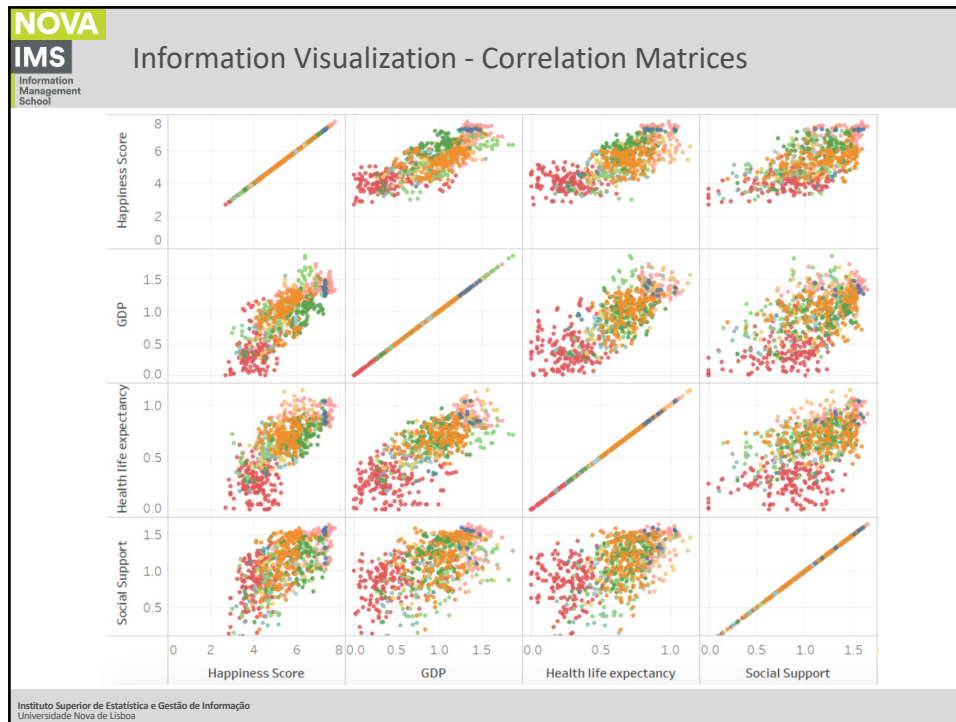
NOVA
IMS
Information Management School

Information Visualization Correlation Matrices

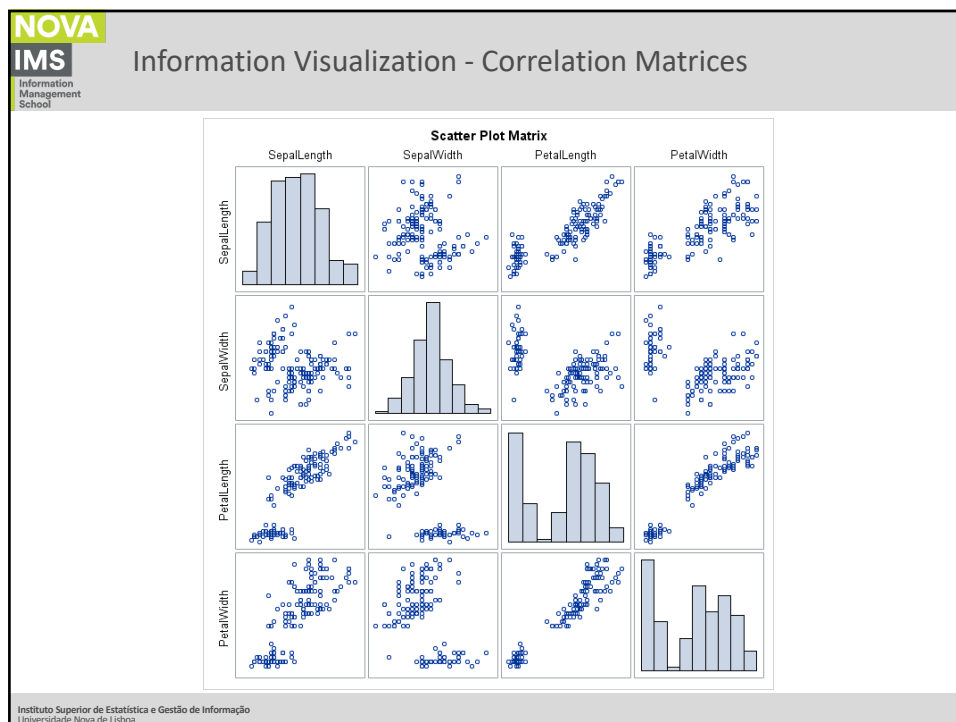
FMI see: <https://builtin.com/data-science/correlation-matrix>

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

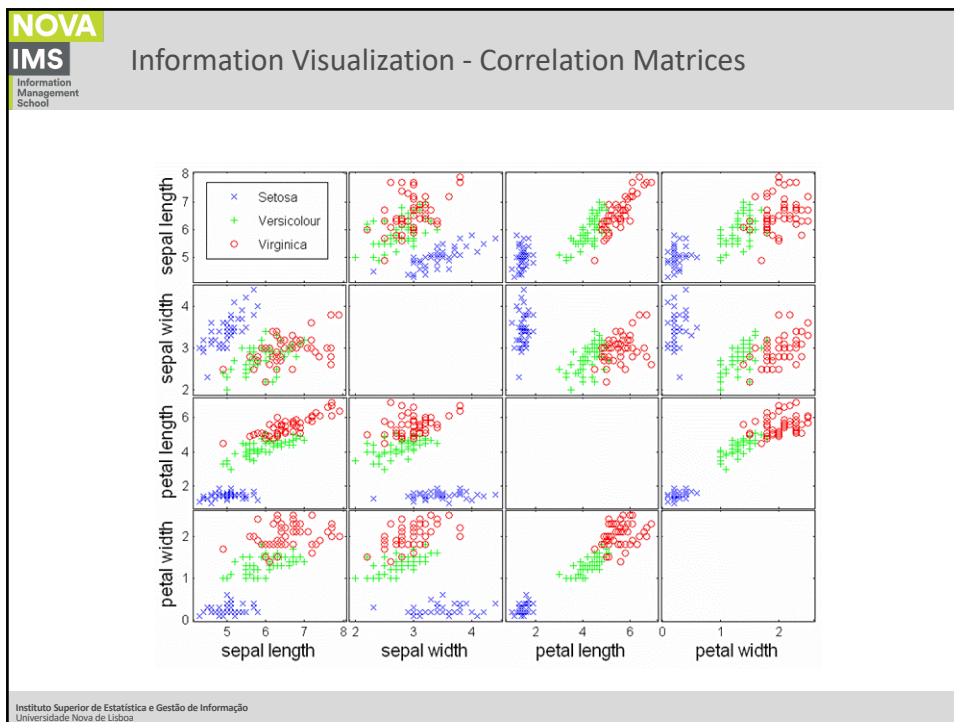
96



97



98



99

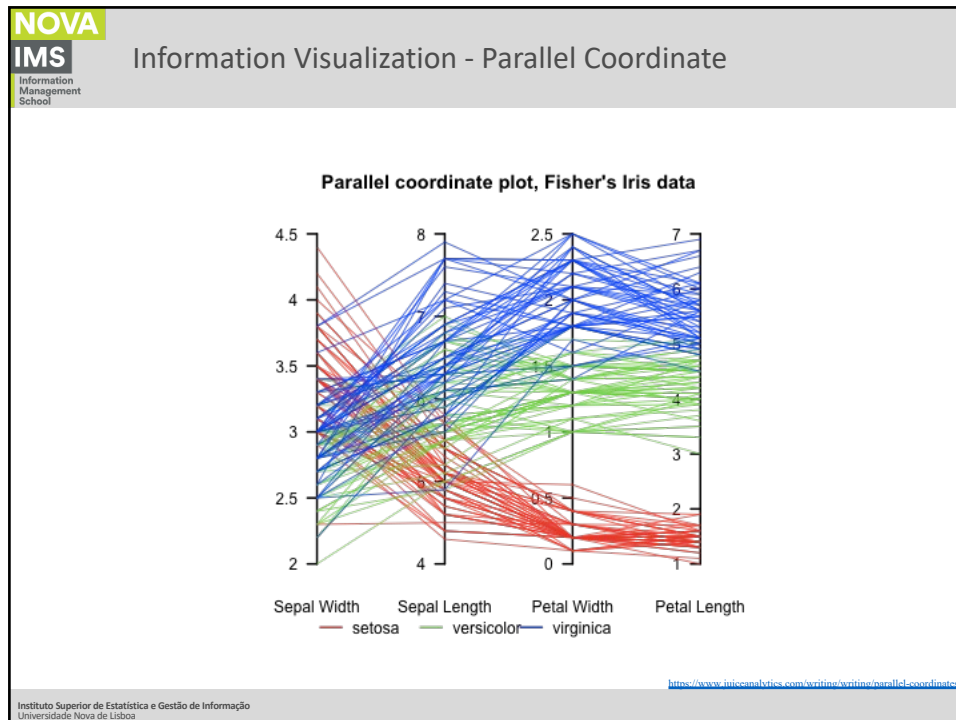
NOVA
IMS
Information Management School

Information Visualization Parallel Coordinates

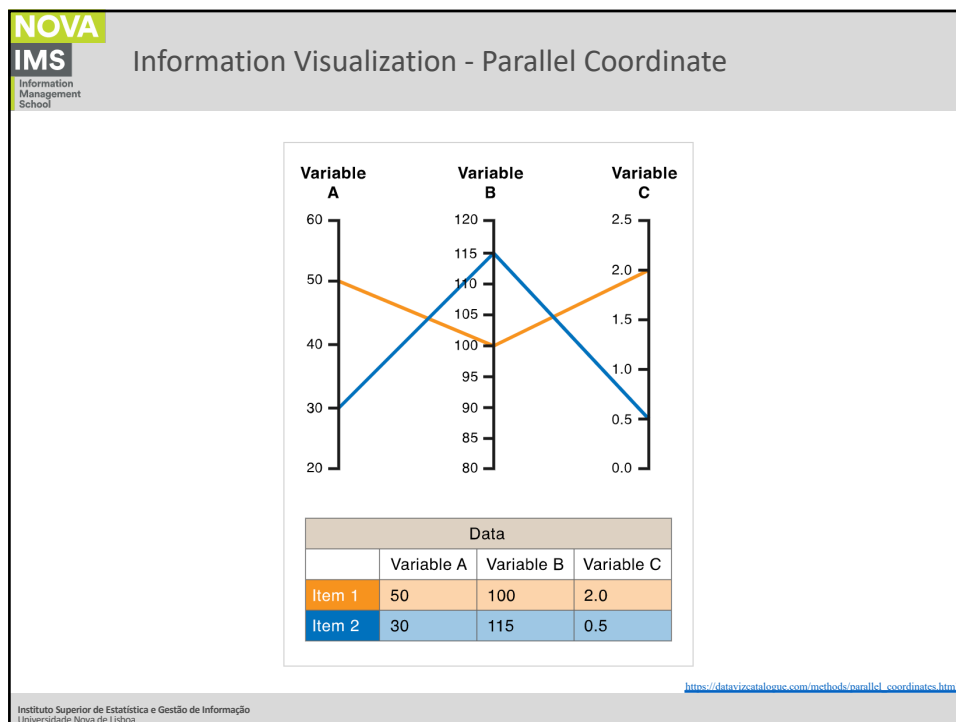
FMI see: <https://www.juiceanalytics.com/writing/writing/parallel-coordinates>

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

100



101



102

NOVA
IMS
Information Management School

Information Visualization Small Multiples

- A small multiple is a series of similar graphs using the same scale + axes, allowing them to be easily compared.
- It uses multiple views to show different partitions of a dataset.

FMI see: <https://www.juiceanalytics.com/writing/better-know-visualization-small-multiples>

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

UNICIS

ASES

Schools

103

NOVA
IMS
Information Management School

Information Visualization – Small Multiples

THE TRILOGY METER

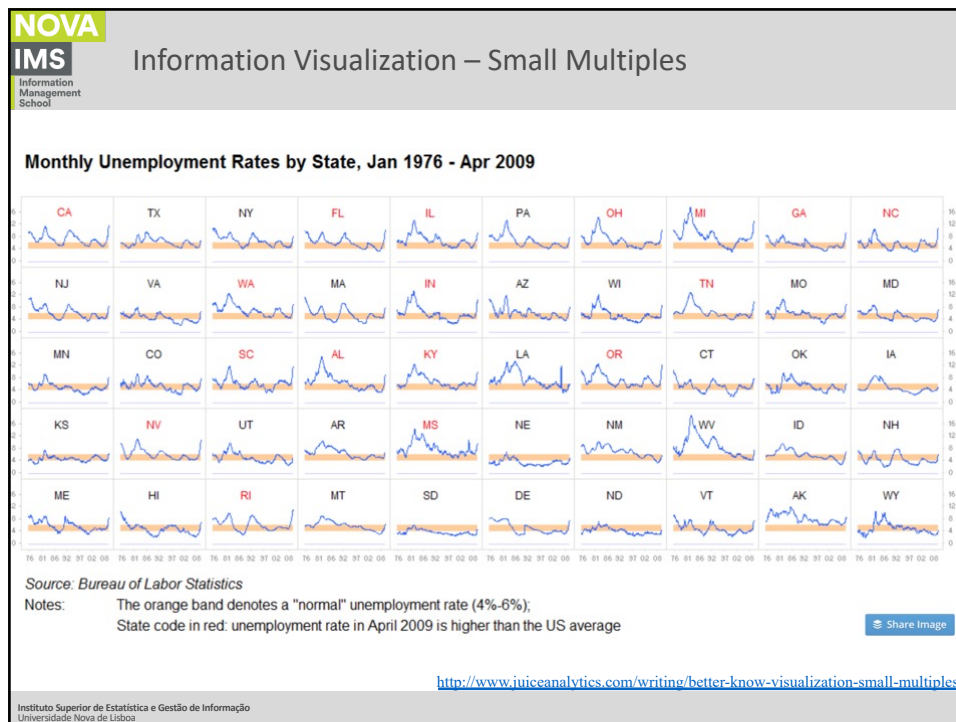
#1 In A Series of Pop Cultural Charts

DANMETH.COM

<http://www.juiceanalytics.com/writing/better-know-visualization-small-multiples>

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

104



105

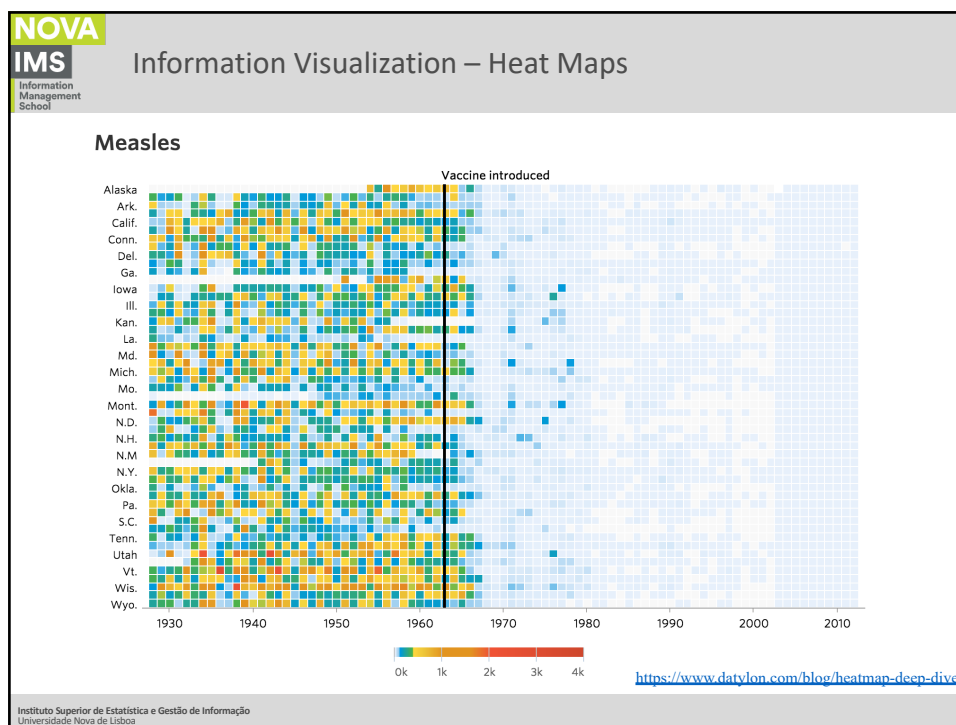
NOVA
IMS
Information Management School

Information Visualization Heat Maps

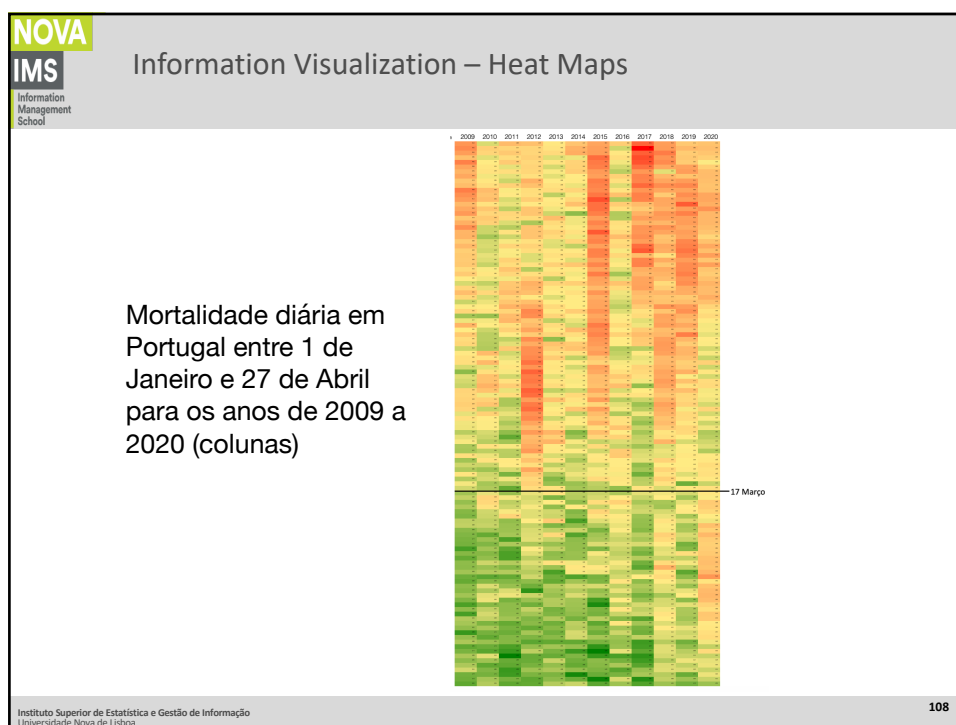
FMI see: <https://www.datylon.com/blog/heatmap-deep-dive>

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

106

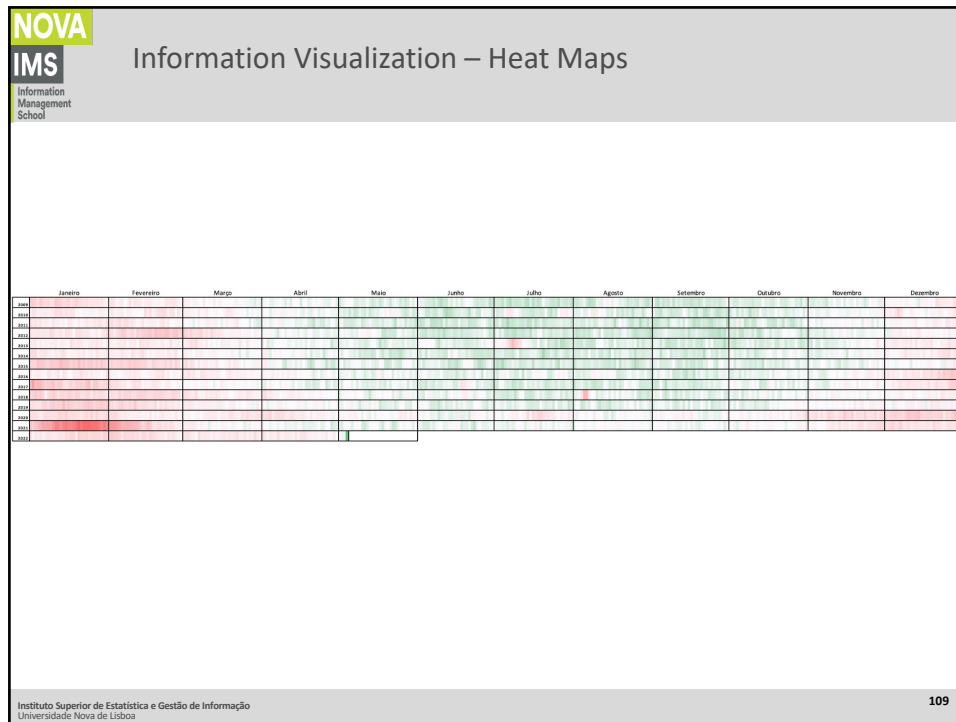


107

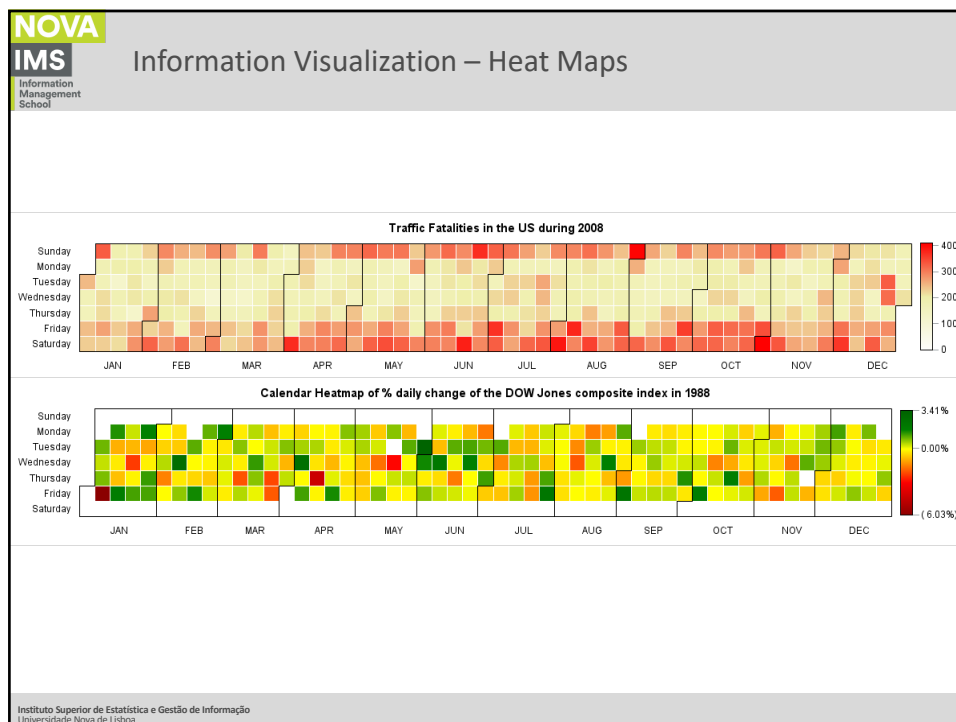


108

108



109



110

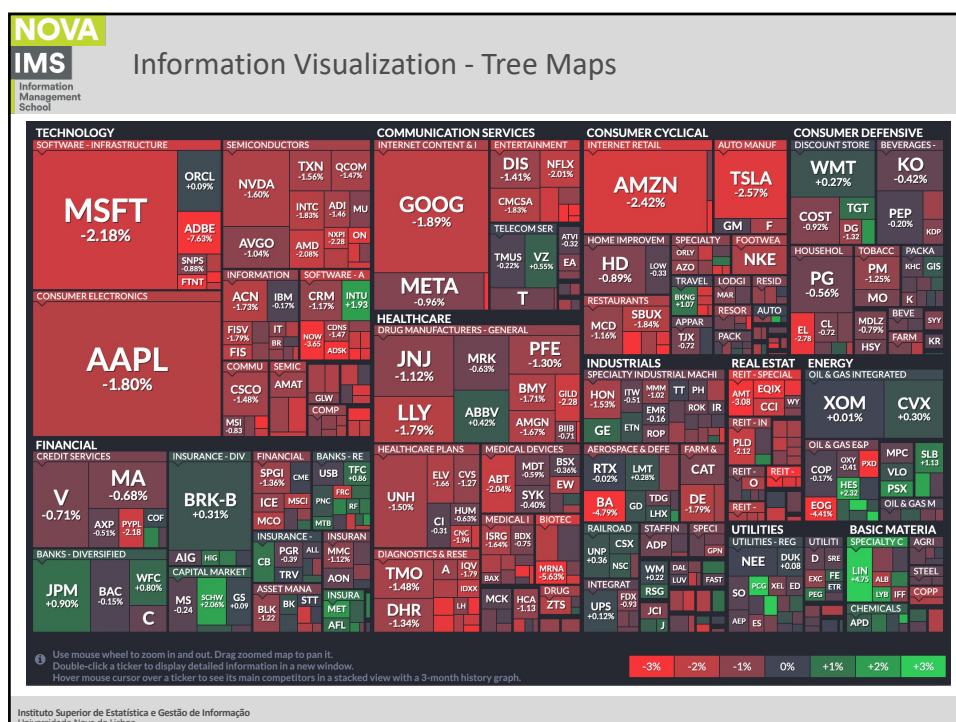
NOVA
IMS
Information Management School

Information Visualization Tree Maps

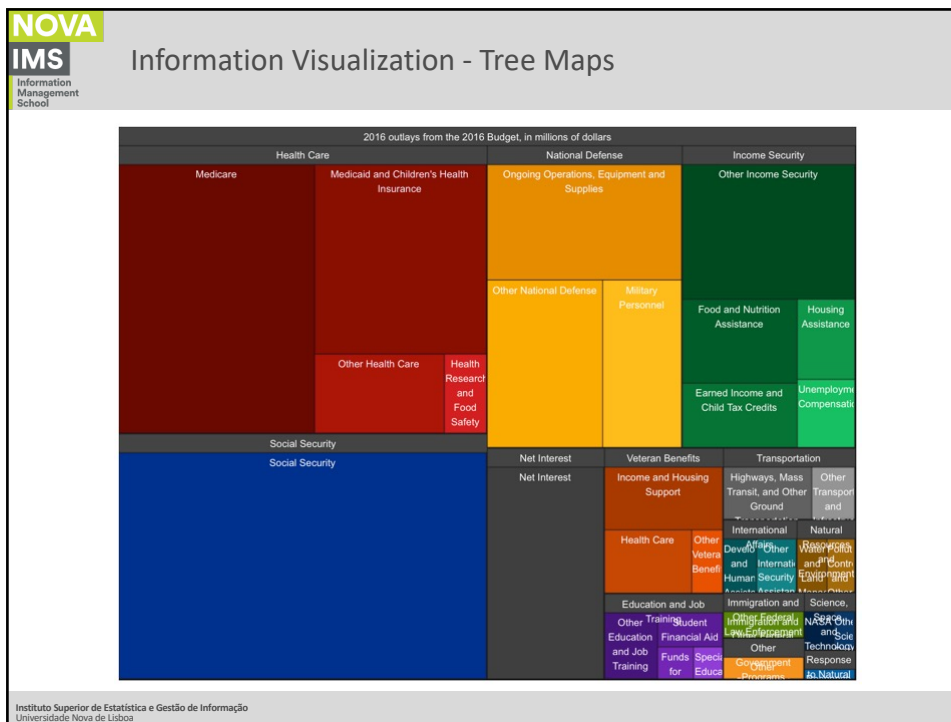
FMI see: <https://www.nnoup.com/articles/treemaps/>

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

111



112



113



114

NOVA
IMS
Information Management School

Information Visualization Linked Views

FMI see: https://courses.ischool.berkeley.edu/f24/105/readings/Baldonado_MultipleViews_AV100.pdf

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

UNICIS

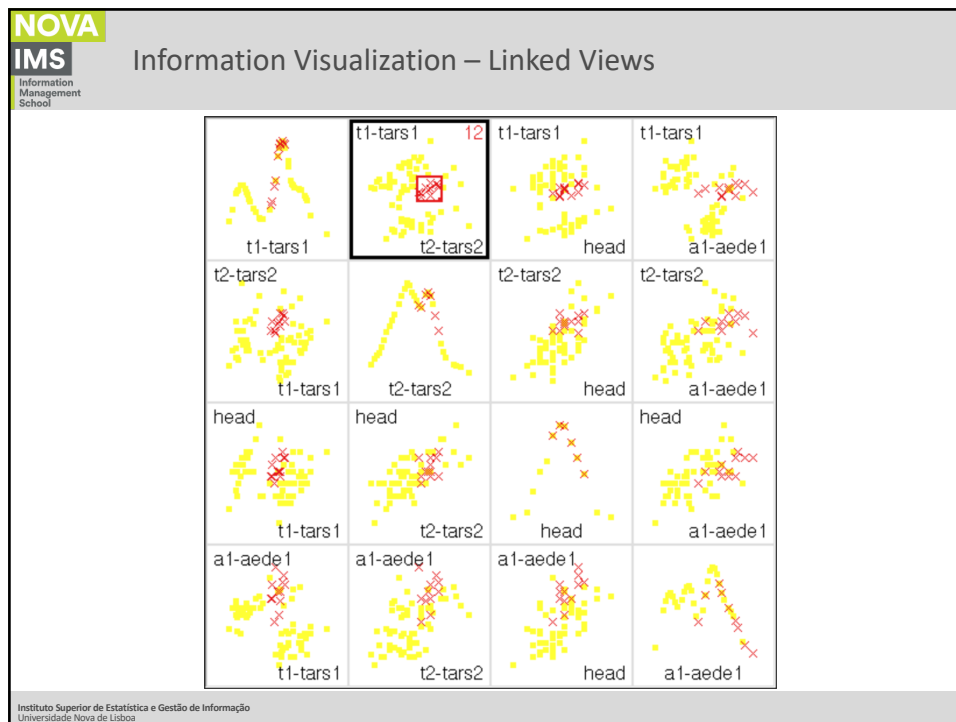
115

NOVA
IMS
Information Management School

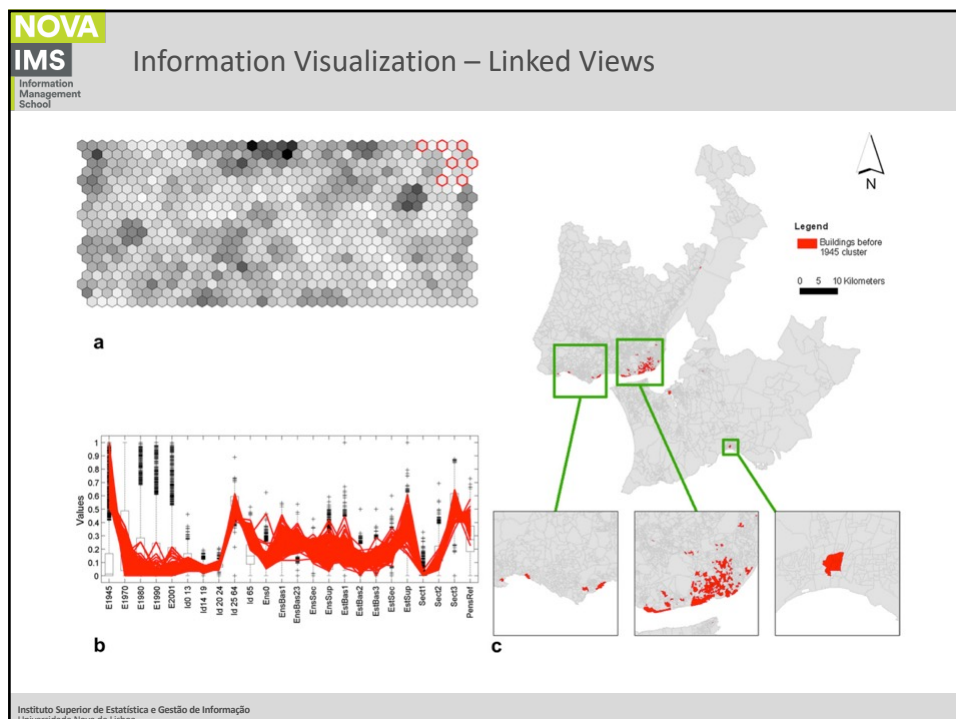
Information Visualization – Linked Views

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

116



117



118

NOVA
IMS
Information Management School

Geo-Visualization

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

UNICIS

119

NOVA
IMS
Information Management School

Information Visualization - Bubble Graphics

Travel restrictions in response to covid-19

At March 16th 2020, 11am GMT

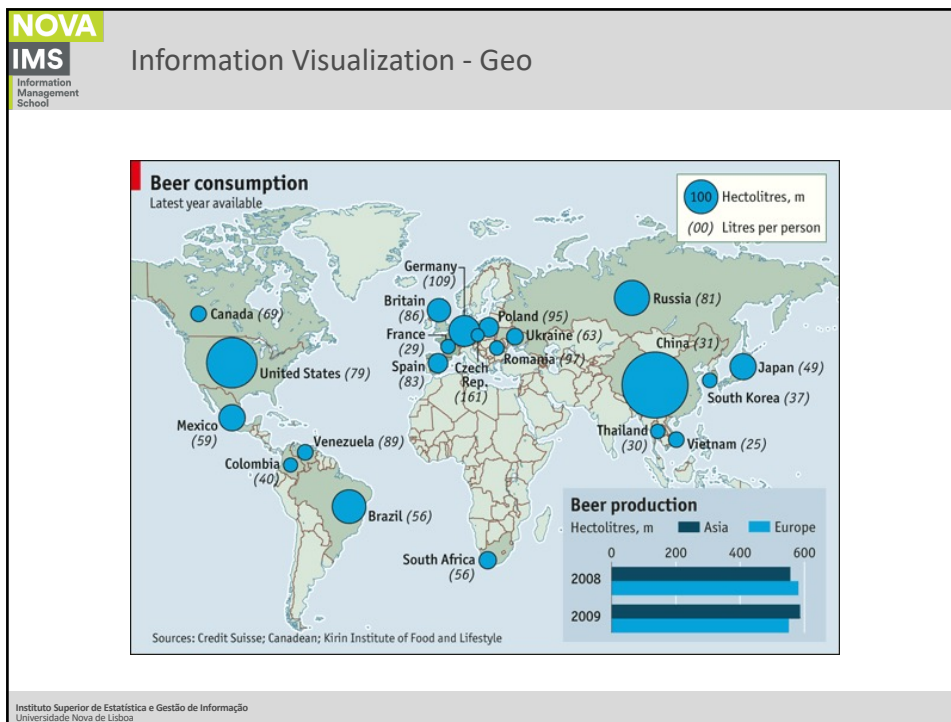
- Quarantine
- Travel ban or flights suspended
- Travel ban and quarantine
- Other

Sources: Mayer Brown; press reports

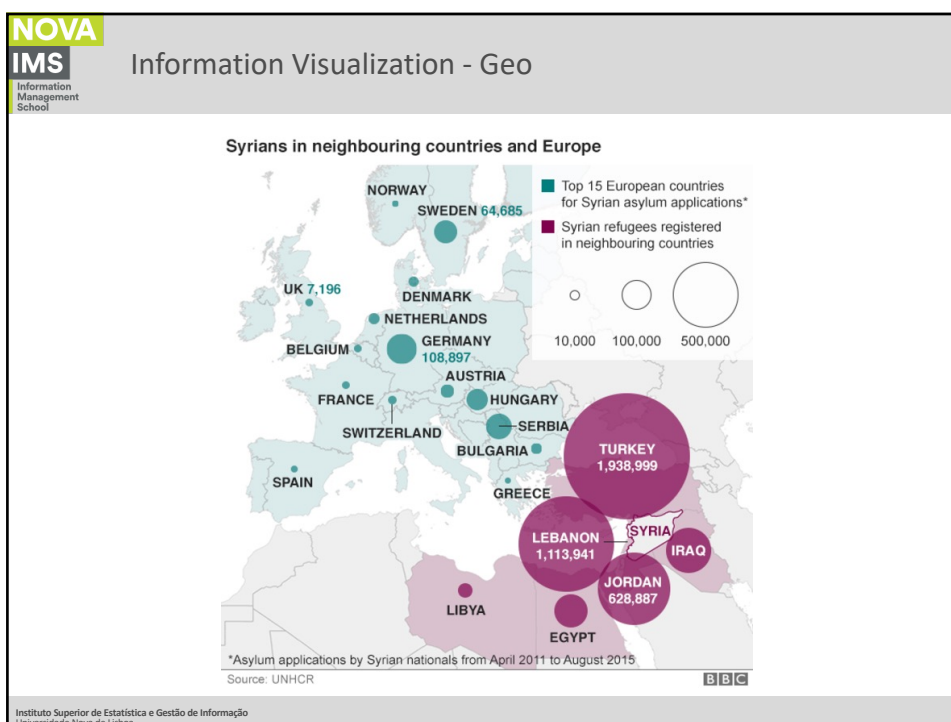
The Economist

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

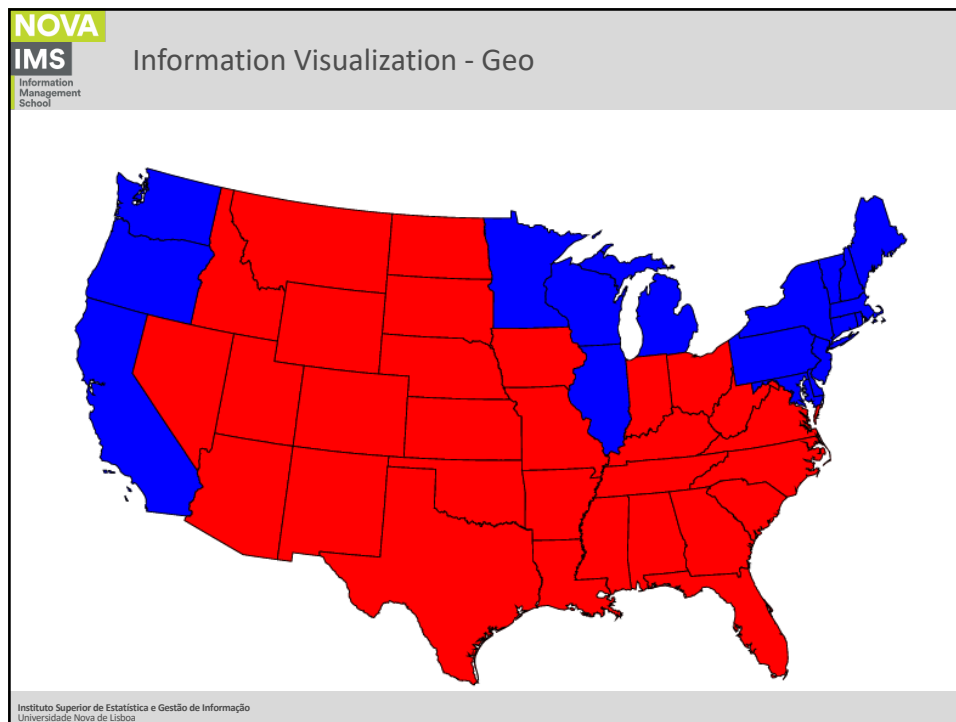
120



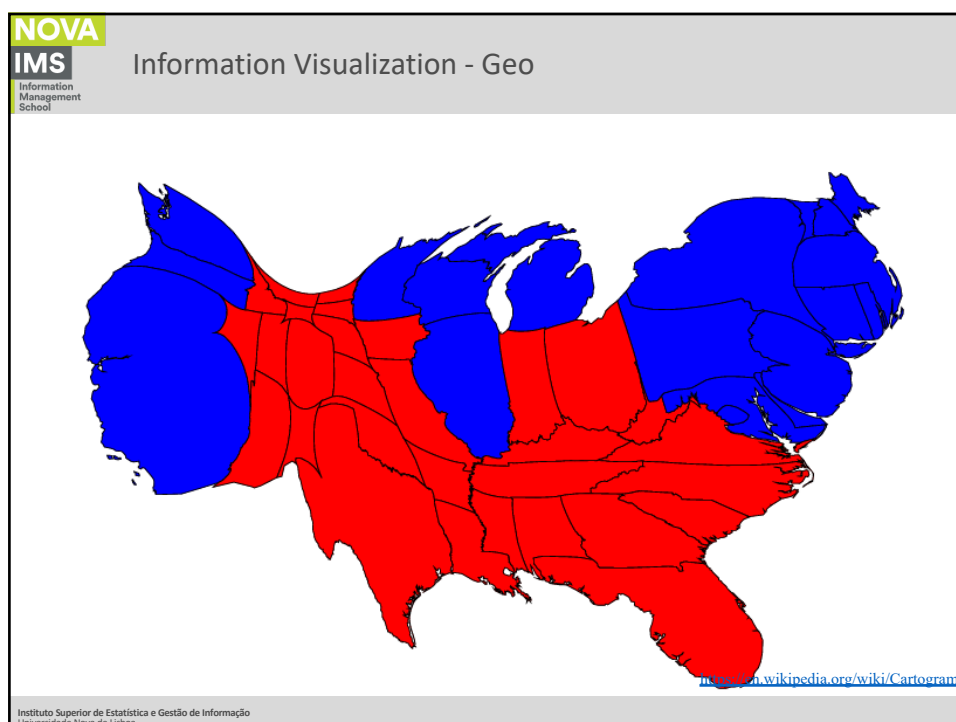
121



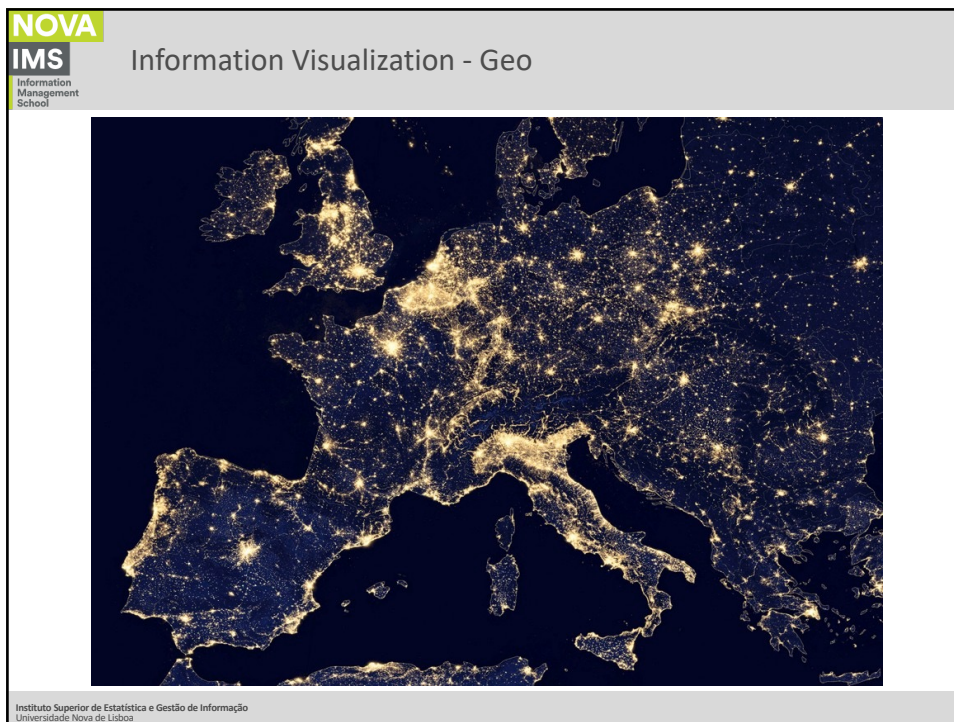
122



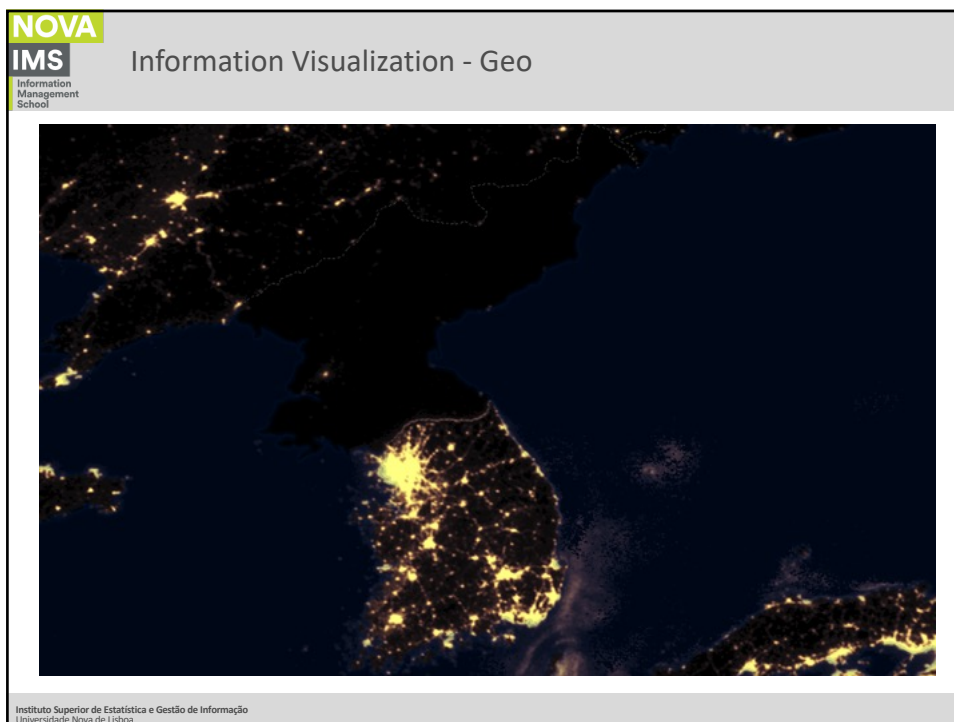
123



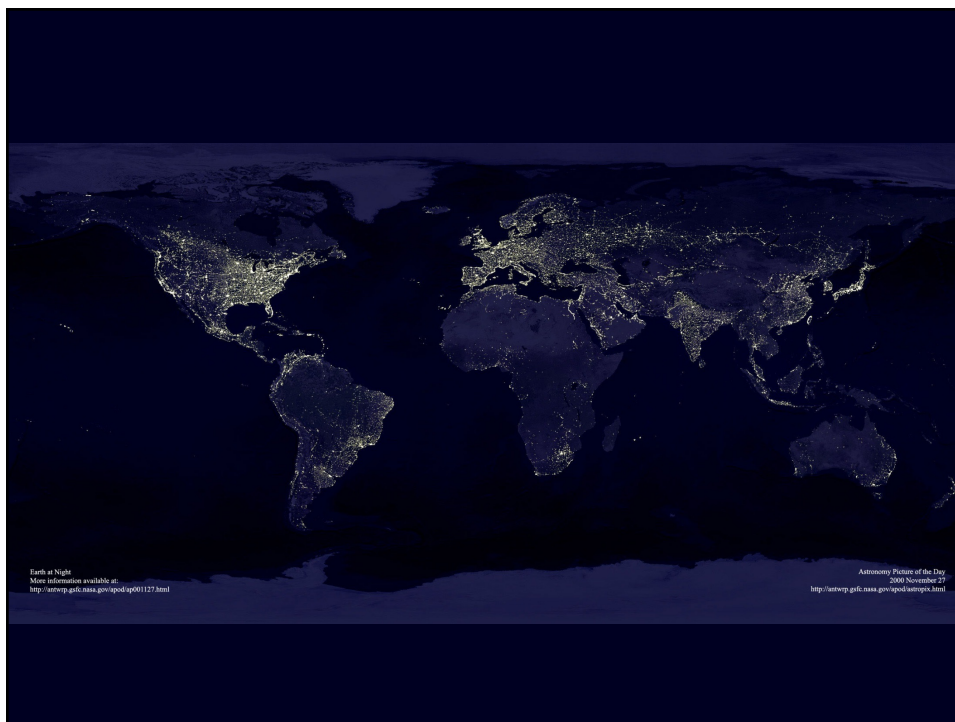
124



125



126



127



128